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NR/L2/ELP/27716

Module 01

Design manual – clearances

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1 Purpose

The purpose of this module is to define a methodology to be used when assessing electrical and mechanical clearances associated with overhead electrification (including pantographs).

Following the module will allow designers to consistently:

i) produce a minimal viable product (MVP) clearance assessment

ii) funders with a list of opportunities in which to invest;

iii) develop evidence-based designs; and

iv) reduce the embedded carbon content of new electrification by avoiding unnecessary overline structure reconstruction.

2 Scope

This standard applies to designers, sponsors and Designated Project Engineers (DPEs) of schemes for new or modified overhead electrification. It specifies the requirements for assessing and quantifying the electrical and mechanical clearances to be met by the electrification design.

3 Legislation

This standard allows the user to easily demonstrate compliance with the following legislation when applying the electrical clearance assessments:

1. The Health and Safety at Work Act 1974
2. The Construction (Design and Management) Regulations 2015 (CDM)
3. The Electricity at Work Regulations 1989 (EaWR)
4. The Management of Health and Safety at Work Regulations 1999 (MHSWR)
5. Common Safety Methods for Risk Evaluation and Assessment 402/2313[1] (CSM RA)
6. The Railways and Other Guided Transport Systems (Safety) Regulations 2006 (ROGS)
7. The Railways (Interoperability) Regulations 2011.

This standard also takes cognisance of:

1. The Control of Electromagnetic Fields at Work Regulations 2016 (CoEMFaW)
2. Electrical Safety, Quality and Continuity Regulations (EQSCR) 2002
3. Land Reform (Scotland) Act 2003
4. The Railway (Miscellaneous Provisions) Regulations 1997
5. The Occupiers' Liability Act 1984
6. The Highways Act 1980
7. New Roads & Street Works Act 1991.

As stated in the ORR 'Common safety method for risk evaluation and assessment – guidance on the application of commission regulation 402/2013',

> *The CSM RA, ROGS, MHSWR and CDM Regulation requirements seek to achieve the same result: a robust risk assessment and controls to maintain or reduce risk. ORR therefore considers it to be unnecessary for duty holders to produce separate risk assessment and evaluation processes to comply with domestic and European requirements. Compliance with the CSM RA should simultaneously deliver compliance with regulation 19 ROGS and regulation 3 of MHSWR the change and impact on other interfaces, as the purpose of the CSM RA is to deliver a thorough and competent risk assessment process.*

Therefore, module 3 of this manual is primarily focused on the requirements of CSM RA with respect to the design specification contained within this standard.

[1] Commission Regulation 402/2013 became retained and corrected EU law on 1 January 2021 following the end of the transition period. This retained legislation was amended by Regulation 16 of the Rail Safety (EU Exit) (Amendment etc.) Regulations 2019 (S.I. 2019/837) and continues to apply in Great Britain.

4 Clearances

4.1.1.1 Requirements

The insulation rating or separation distance for the clearances in Figure 1 shall be assessed. Where relevant, the clearances shall be assessed for their static and dynamic states.

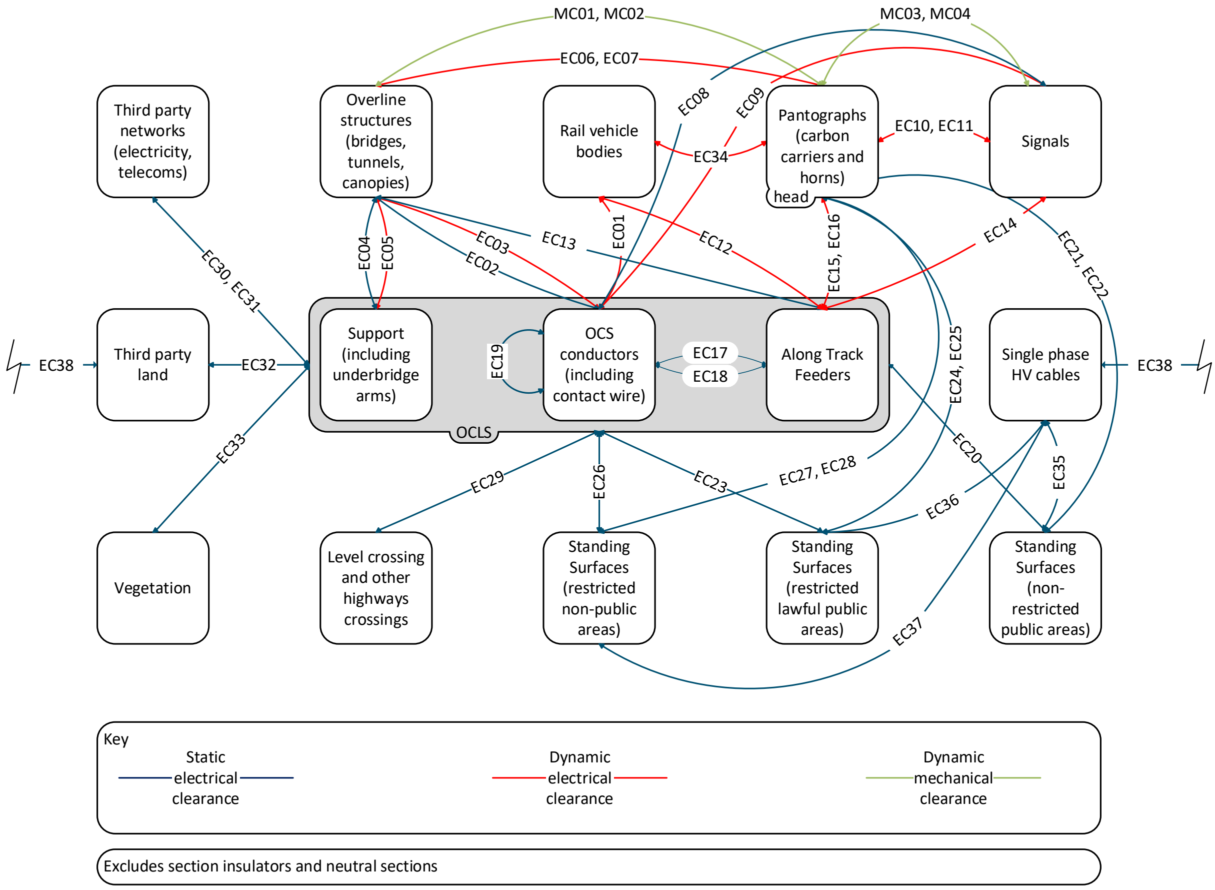

Figure 1 – Electrical and mechanical clearances

These clearances are listed in Table 1 and Table 2.

	From	Victim	Type
EC01	OCL conductors (including contact wire)	Rail vehicle bodies	Dynamic
EC02	OCL conductors (including contact wire)	Overline structures (bridges, tunnels, canopies)	Static
EC03	OCL conductors (including contact wire)	Overline structures (bridges, tunnels, canopies)	Dynamic
EC04	Support (including underbridge arms)	Overline structures (bridges, tunnels, canopies)	Static
EC05	Support (including underbridge arms)	Overline structures (bridges, tunnels, canopies)	Dynamic
EC06	Pantographs – carbon carriers	Overline structures (bridges, tunnels, canopies)	Dynamic
EC07	Pantographs – horn	Overline structures (bridges, tunnels, canopies)	Dynamic
EC08	OCL conductors (including contact wire)	Signals	Static
EC09	OCL conductors (including contact wire)	Signals	Dynamic
EC10	Pantographs – carbon carriers	Signals	Dynamic
EC11	Pantographs – horn	Signals	Dynamic
EC12	Along-track feeders	Rail vehicle bodies	Dynamic
EC13	Along-track feeders	Overline structures (bridges, tunnels, canopies)	Static
EC14	Along-track feeders	Signals	Static
EC15	Along-track feeders	Pantographs – horn	Dynamic
EC16	Along-track feeders	Pantographs – carbon carriers	Dynamic
EC17	OCL conductors (including contact wire)	Along-track feeders	Static
EC18	OCL conductors (including contact wire)	Along-track feeders	Static – vertical working clearance
EC19	OCL conductors (including contact wire)	OCL conductors (including contact wire)	Static

	From	**Victim**	**Type**
EC20	OCLS	Standing surfaces (non-restricted public areas)	Static
EC21	Pantographs – horn	Standing surfaces (non-restricted public areas)	Static *
EC22	Pantographs – carbon carriers	Standing surfaces (non-restricted public areas)	Static *
EC23	OCLS	Standing surfaces (restricted lawful public areas)	Static
EC24	Pantographs – horn	Standing surfaces (restricted lawful public areas)	Static *
EC25	Pantographs – carbon carriers	Standing surfaces (restricted lawful public areas)	Static *
EC26	OCLS	Standing surfaces (restricted non-public areas)	Static
EC27	Pantographs – horn	Standing surfaces (restricted non-public areas)	Static *
EC28	Pantographs – carbon carriers	Standing surfaces (restricted non-public areas)	Static *
EC29	OCLS	Level crossing and other highways crossings	Static
EC30	OCLS	Third party networks - electricity	Static
EC31	OCLS	Third party networks - telecoms	Static
EC32	OCLS	Third party land	Static
EC33	OCLS	Vegetation	Static
EC34	Pantographs – horn	Rail vehicle bodies	
EC35	Single phase HV cables	Standing surfaces (unrestricted public areas)	Static
EC36	Single phase HV cables	Standing surfaces (restricted public areas)	Static
EC37	Single phase HV cables	Standing surfaces (restricted non-public areas)	Static
EC38	Single phase HV cables	Third party land	Static

Table 1 – Electrical clearances

* Strictly these clearances are dynamic, however these are assessed as static clearances as the presence of a rail vehicle moving at speed is a greater risk.

Note EC34 is managed by GMRT2111, and is therefore outside the scope of infrastructure design considerations.

	From	Victim	Type
MC01	Pantographs – horn	Overline structures (bridges, tunnels, canopies)	Dynamic
MC02	Pantographs – carbon carriers	Overline structures (bridges, tunnels, canopies)	Dynamic
MC03	Pantographs – horn	Signals	Dynamic
MC04	Pantographs – carbon carriers	Signals	Dynamic
MC05	Pantographs – carbon carriers	Support (including underbridge arm)	Dynamic

Table 2 – Mechanical clearances

4.1.1.2 Rationale

Most of the legislation identified provide the requirements for risk assessment and prevention of danger. The Electricity at Work Regulation 7 provides explicit requirements. This states:

> *All conductors in a system which may give rise to danger shall either -*
>
> *(a) be suitably covered with insulating material and as necessary protected so as to prevent, so far as is reasonably practicable, danger; or*
>
> *(b) have such precautions taken in respect of them (including, where appropriate, their being suitably placed) as will prevent, so far as is reasonably practicable, danger.*

The HSE Guidance note HSR25, para 110, states:

> *Some form of basic insulation, or physical separation, of conductors in a system is necessary for the system to function. That functional minimum, however, may not be sufficient to comply with the requirements of regulation 7. Factors which must be taken into account are:*
>
> *(a) the nature and severity of the probable danger;*
>
> *(b) the functions to be performed by the equipment;*
>
> *(c) the location of the equipment, its environment and the conditions to which it will be subjected;*
>
> *(d) any work which is likely to be performed upon, with or near the equipment.*

This standard provides designers with clarity on how clearances should be calculated and assessed and what physical separation is considered sufficient to mitigate danger.

4.1.1.3 Guidance

All clearances should be calculated and assessed to evidence compliance with legislation.

This standard provides linkage for OLE designers to apply and justify the application of appropriate best practices, adherence to Euronorms or other codes of practice to electrical and mechanical clearances. It recognised that it might not be possible to fully comply with all standards due to conflicts between them. Equally it might not be reasonably practicable to comply due to the disproportionate extent of the works required to achieve the preferred clearances. This standard therefore provides a hierarchy of design options along with safety justification to allow OLE designers to focus on producing designs that balances these requirements and one that is aligned to the minimal viable product.

5 Electrical insulation categories

5.1.1.1 Requirements

When specified by this standard, electrical clearances shall be classified to identify the electrical insulation categories in accordance with Table 3.

	Electrical insulation category	Rated impulse voltage (U_{Ni})	
		Minimum	Maximum
1	Reinforced	320 kV	or greater
2	Basic	200 kV	319 kV
3	Functional (static)	145 kV	199 kV
4	Functional (dynamic)	108 kV	145 kV
5	Basic (VCC)	200 kV	319 kV
6	Functional (stochastic)	Less than 108 kV	

Table 3 – Electrical clearance (insulation coordination) classification

Site-specific risk assessments shall be carried out for electrical clearances rated as Functional (stochastic).

5.1.1.2 Rationale

These categories align with BS EN 50124-1, and incorporate the typical values specified in BS EN 50119.

Table 3 also includes a category entitled Basic (VCC). This category provides the same electrical performance as basic insulation, but uses surge arresters and may additionally include supplementary insulation or control measures.

The duration for which electrical clearances exist is important. BS EN 50119 subclause 5.1.3 states

> *"Different clearances for static and dynamic cases are justifiable by probabilistic determinations taking account of duration. For example, it is improbable that an over-voltage surge will occur at the same moment that a pantograph passes a narrow part of a tunnel. For this "dynamic" or temporary case, the use of a dynamic clearance is justified."*

5.1.1.3 Guidance

Module 5 details the minimum physical distances required to provide the various electrical insulation categories. The dimensions that relate specifically to OCLS and pantograph clearances are as a result of testing carried out by Network Rail in accordance with BS EN 50124-1 and BS EN 60060-1.

BS EN 50122-1 subclause 5.1.3 table 1 contains a value for providing a rounded electrical clearance in air *d*. It also permits the use of BS EN 50124-1 and surge arresters to provide alternative values for rounded electrical clearance in air *d*. This approach is used by this standard.

When determining clearances for locations that could have people present, anthropometric and ergonomic considerations dominate, and the electrical insulating performance of an air gap is less important. These clearances tend to be specified as a physical distance from a standing surface.

6 Design

6.1 Option development

6.1.1.1 Requirements

Clearances shall be assessed during the development phase of the project to determine if route clearance works are required to be included in the project scope.

Whenever route clearance works are required, the design options shall be evaluated in accordance with the minimum viable product (MVP) principles. The MVP design option shall include

- the minimum input parameters contained within module 4 of this manual;
- probabilistic pantograph gauging for arched overline structures;
- insulated pantograph horns only operation;
- standard rail vehicle gauge profile; and
- zero track maintenance lift allowance (TMLA);

and

- not limited by contact wire gradient.

This option shall be used to inform the sponsor of the MVP design option.

Where the MVP design option provides insufficient clearances for basic VCC then additional design options with civils works shall be developed. This should include

- track lower designs shall be designed in accordance with NR/L2/TRK/2102 clause 6.8.2. These designs shall utilise the control measures listed to limit the depth of track lower.

and / or

- designs for new and reconstructed overline structures in accordance with NR/L3/CIV/020.

These MVP options shall be developed to include

- the minimum input parameters contained within module 4 of this standard,
- probabilistic pantograph gauging for arched overline structures,
- insulated pantograph horns only operation,
- standard rail vehicle gauge profile,
- basic VCC clearances,

and

- not limited by contact wire gradient.

The designer shall also identify options to inform the project funder of opportunities to optimise the design.

These options could include:

- installing slab track to remove track fixity maintenance requirements;
- providing larger OCLS construction and maintenance tolerances;
- providing a (non-zero) TMLA to reduce the level of precision required for track maintenance activities;

NOTE: *NR/L3/TRK/3417 (Specification, installation and maintenance of Managed Track Position) provides controls so that the track is tamped as per the track design (i.e. lifting within the track fixity limits). Therefore there is no need to provide TMLA when a managed track position level 2 is specified.*

- providing track lateral resistance plates;
- additional clearance for legacy rail vehicles (i.e. steam trains);
- providing additional mechanical clearance to allow future modification to the structure (temporary or permanent) with reduced service impact;
- providing clearances for conductive pantograph horns;
- reducing the steepness of the contact wire gradient;
- providing an allowance for conductor rise due to loss of mass to reduce maintenance requirements;
- increasing clearances.

For each of the designs, the designer shall specify all of the input parameters used to establish the clearances for the assessment, and when these vary from the values specified in module 4.

The Sponsor shall instruct the DPE which opportunities are to be incorporated into the design.

The works shall be assessed for the final configuration of the modification and all stageworks.

6.1.1.2 Rationale

The Construction (Design and Management) Regulations 2015 require designers to assess the practicability of all the associated stage works that may be required to achieve final implementation of the designs.

6.1.1.3 Guidance

The following standards describe the process for the DPE and the sponsor to assess the MVP and opportunities for funding:

- NR/L2/P3M/201 Project Acceleration in a Controlled Environment (PACE); and
- NR/L2/P3M/221 Project Acceleration in a Controlled Environment (PACE) – Manage scope.

6.2 Detailed design

6.2.1.1 Requirements

The detailed design documentation shall include:

- all input parameters used to undertake the clearance assessment;
- all design decision logs;
- optional value opportunities investigated and instructions from funders;
- details of any mitigation measures to be applied;
- the minimum dimension for each clearance assessed; and
- the issue of this standard used to develop the design.

For each bridge arm location, a cross-section drawing shall be produced. These drawings shall contain the following as a minimum:

- track datum position (and values specified for track datum plate or plates);
- design contact wire height;
- minimum contact wire height;
- maximum contact wire height;
- minimum clearance from top of support arm to overline structure (static);
- minimum clearance from conductors (including cover if included in the design) to overline structure (static);
- minimum clearances from the static pantograph outline to overline structure; and
- bridge soffit height.

The works shall be assessed for the final configuration of the modification and all stageworks.

6.2.1.2 Rationale

The ORR guidance on electrical clearances states *“it is not acceptable to argue that a measure is not necessary to ensure safety SFAIRP on the basis of excessive cost if that measure could and should have been identified at an earlier point in the project when its implementation would have been required”.* It is therefore a requirement that all stageworks are assessed as part of the design.

6.3 Modifications to existing electrified railways

6.3.1.1 Requirements

When modifying an existing installation, the design shall be assessed by either of the following two methods:

- compliance with the clearances in this standard,

  or

- demonstration that the cost of achieving the clearances set out in this standard are grossly disproportionate; and
  - distances have been maximised so far as is reasonably practicable; and
  - the existing clearance from any standing surface to live parts of the OCLS and pantograph is a minimum of 2.75 m; and
  - the modification does not introduce any new electrical clearance hazards or increase any existing electrical risks.

The modification works shall be assessed for the final configuration of the modification and stageworks.

Historically “half an insulator” was permitted to be positioned over the station platform edge. While the design rule no longer exists, the arrangement is accepted and “half an insulator” installations can remain until the OCLS structure is renewed. The “half an insulator” arrangement is also permitted in cases where a platform is extended, and an existing headspan structure is present.

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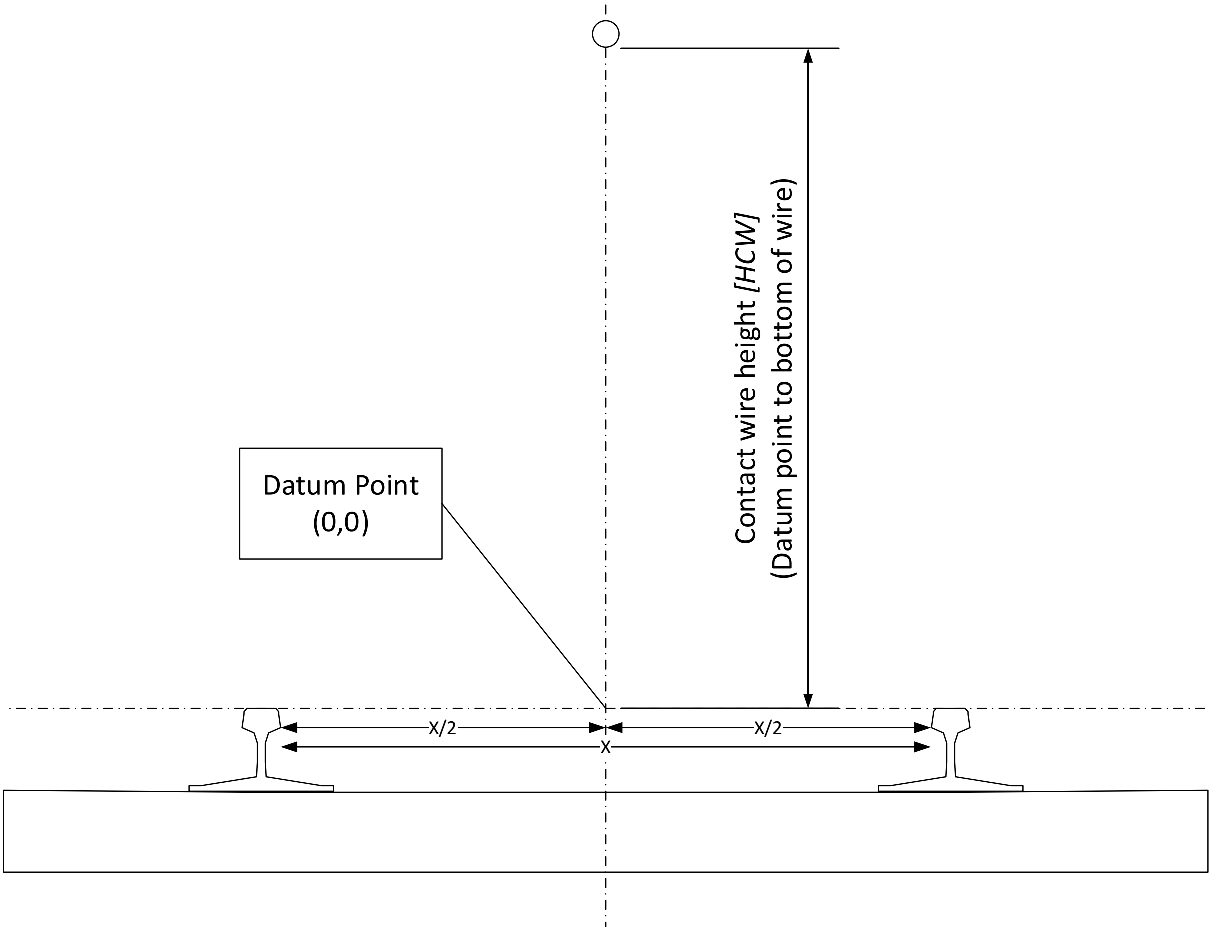
7 Contact wire height

7.1.1.1 Requirements

The contact wire height shall be measured relative to the track datum point in accordance with Figure 2.



X = track gauge

Figure 2 – Contact wire height

The minimum contact wire height [*HCWmin*] shall be either:

- 4165 mm,

  or
- 4040 mm for auto-tensioned equipment with Basic (VCC) insulation.

Locations where the contact wire height [*HCWmin*] is less than 4165 mm shall be identified:

as a residual hazard on isolation diagrams (see NR/L2/ELP/27550),

and

- with onsite signage using RTE 6029 warning sign (see NR/L2/ELP/21131), and

- on the route list of locations where the OCLS could be affected by emissions from a steam locomotive chimney,

  and

- in the route hazard register.

Locations with contact wire height [HCW_{min}] less than 4165 mm shall not have any road rail access points (RRAPs) located underneath.

7.1.1.2 Rationale

This requirement is based on GLRT1210 issue 3 and RIS-3440-TOM.

The signage will provide a reminder to train drivers exiting their cab, and to on-track plant operators.

7.1.1.3 Guidance

NR/L2/ELP/27550 Module 1A, issue 2, details requirement for residual hazards on isolation diagrams.

NR/L2/ELP/21131 issue 4 details warning sign RTE 6029 Warning sign: Exceptionally low wire heights.

RIS-3440-TOM issue 2 requires a list of locations where the OCLS could be affected by emissions from steam locomotive chimney to be made available to Railway Undertakings.

NR/L2/RMVP/0200/P301 issue 4 (subclause 10.6) details the minimum contact wire height for RRAPs.

The minimum contact wire height currently in use on Network Rail is 3925 mm [source EHQ/ST/O/003 - Standards for the maintenance of Mark IIIB overhead line equipment].

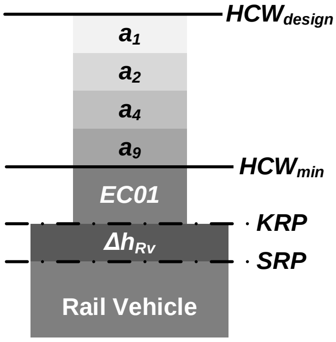
8 Electrical Clearances

8.1 OCLS conductors (including contact wire) to rail vehicles [EC01]

8.1.1.1 Requirements

The electrical clearance between the contact wire and the rail vehicle shall be calculated in accordance with Figure 3.

No additional parameters or allowances shall be used for the MVP design option.



a_1	vertical upwards tolerance of the track
a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_9	contact wire pre-sag
Δh_{Rv}	vertical superelevation

Figure 3 – Electrical clearance diagram [EC01]

The electrical insulation of the clearances shall be assessed using module 5 using parameters from module 4.

The designer shall prioritise achieving the highest electrical insulation category in accordance with Table 4.

	Category				
	1	**2**	**3**	**4**	**5**
EC01	Reinforced	Basic	Functional (dynamic)	Basic (VCC)*	–

* Basic VCC shall only be used with auto tensioned equipment.

Table 4 – Electrical insulation category [EC01]

8.1.1.2 Rationale

GLRT1210, issue 3, subclause 3.1.1.1 requires the contact wire position to be calculated in accordance with BS EN 50119:2020 figure 3. The approach detailed in this standard aligns with this requirement whilst removing some elements not applicable to Network Rail's specific GB requirements.

This calculation methodology allows the consistent assessment of the hazard identified in the bow tie “risk of flashover caused by low contact wire” (see module 3).

8.1.1.3 Guidance

The application of pre-sag is dependent on the OCLS design range.

A usual train stop on a railway line (e.g. in a station) is considered as a temporary case due to the limited duration (see IEC CD 60913:2021).

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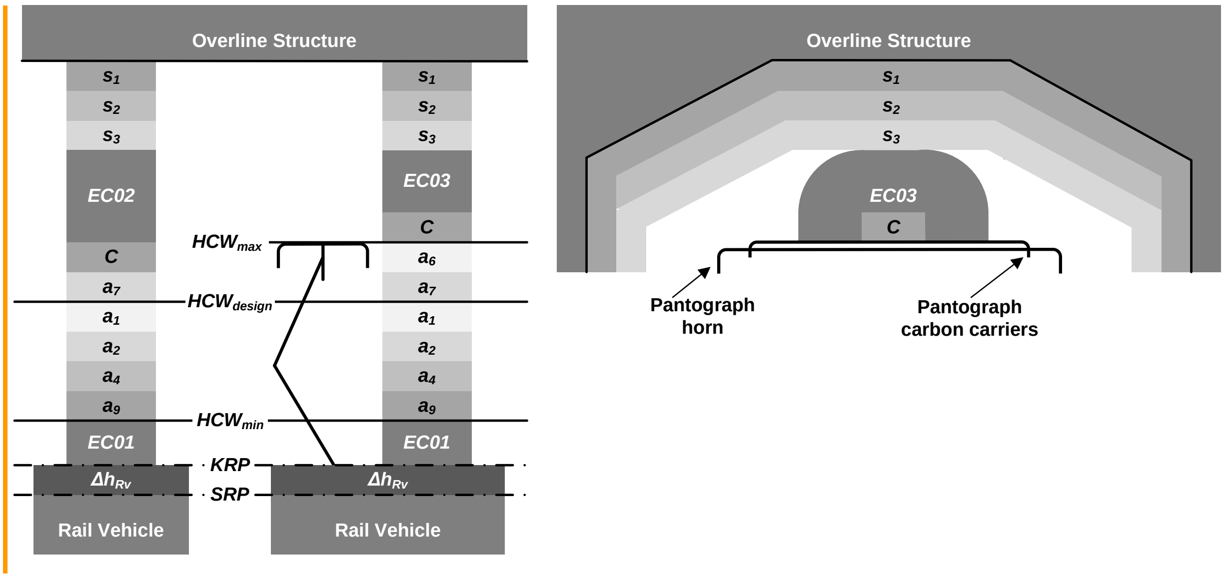
8.2 OCL conductors to overline structures [EC02, EC03]

8.2.1.1 Requirements

The static (EC02) and dynamic (EC03) electrical clearance between the conductors (contact wire and catenary) and overline structures shall be calculated in accordance with Figure 4.

No additional parameters or allowances shall be used for the MVP design option.



a_1	vertical upwards tolerance of the track
a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_6	uplift of the contact wire by the pantograph
a_7	upwards installation tolerance for the conductors
a_9	contact wire pre-sag
s_1	structure measurement tolerance
s_2	structure deflection
s_3	supplementary insulation
Δh_{Rv}	vertical superelevation
C	conductors (contact wire and catenary, or double contact wire plus contact wire cover)

Figure 4 – Electrical clearance [EC02, EC03]

The insulation category of the minimum dimension for EC02 and EC03 shall be assessed using tables in module 5 and parameters from module 4 of this standard.

The designer shall prioritise achieving the highest electrical insulation category in accordance with Table 5 and Table 24 without significant cost.

	Category				
	1	2	3	4	5
EC02	Reinforced	Basic	Functional (static)	Basic (VCC)	Functional (stochastic)
EC03	Reinforced	Basic	Functional (dynamic)	Basic (VCC)	Functional (stochastic)

Table 5 – Electrical insulation category [EC02, EC03]

8.2.1.2 Rationale

This aligns with the requirements in BS EN 50119:2020 figure 3 for calculating wire heights.

This calculation methodology allows the consistent assessment of the hazard identified in the bow tie “risk of flashover to overline structure” (see module 3).

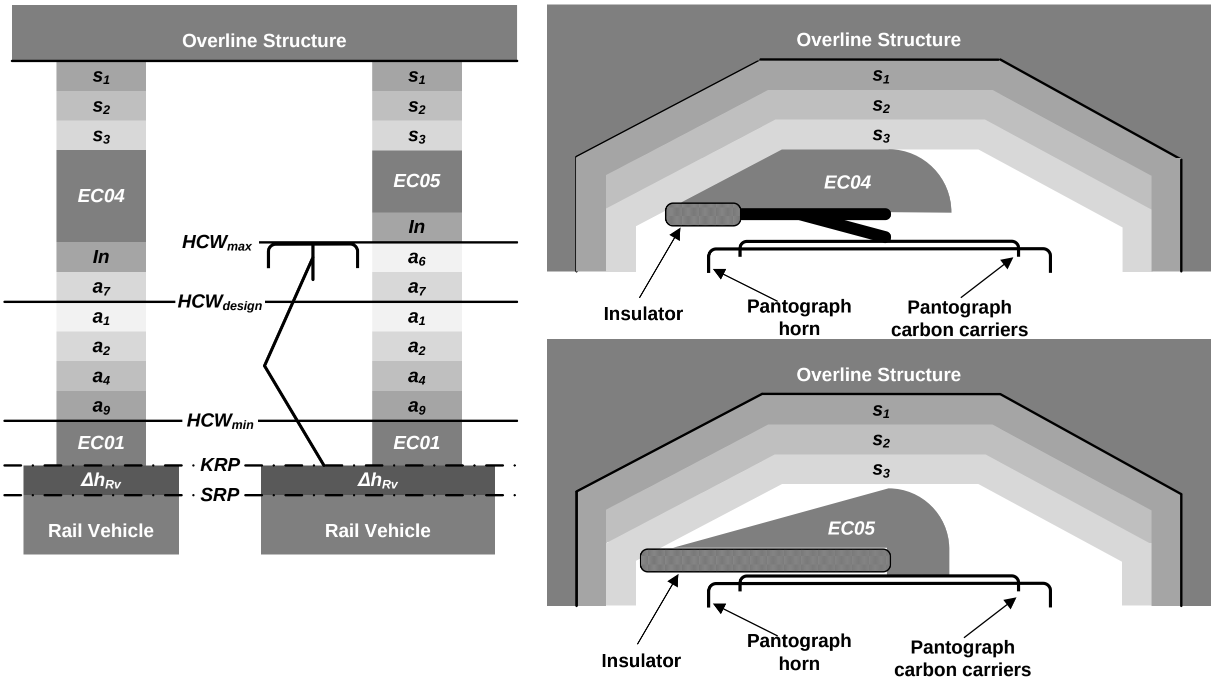
8.3 Support (including underbridge arm) to overline structures [EC04, EC05]

8.3.1.1 Requirements

The clearance between supports not subjected to uplift movement and overline structures (EC04) shall be calculated in accordance with Figure 5.

The clearance between an insulated underbridge arm which is subjected to uplift movement and an overline structure (EC05) shall be calculated in accordance with Figure 5.

No additional parameters or allowances shall be used for the MVP design option.



a_1	vertical upwards tolerance of the track
a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_6	uplift of the contact wire by the pantograph
a_7	upwards installation tolerance for the conductors
a_9	contact wire pre-sag
s_1	structure measurement tolerance
s_2	structure deflection
s_3	insulated coating
Δh_{Rv}	vertical superelevation
In	insulator or bridge arm

Figure 5 – Electrical clearance diagram [EC04, EC05]

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The insulation category of the minimum dimension for EC04 and EC05 shall be assessed using tables in module 5 and parameters from module 4 of this standard.

The designer shall prioritise achieving the highest electrical insulation category in accordance with Table 6 and Table 24 without significant cost.

	Category				
	1	2	3	4	5
EC04	Reinforced	Basic	Functional (static)	Basic (VCC)	Functional (stochastic)
EC05	Reinforced	Basic	Functional (dynamic)	Basic (VCC)	Functional (stochastic)

Table 6 – Electrical insulation category [EC04, EC05]

8.3.1.2 Rationale

This requirement aligns with the requirements in BS EN 50119.

This calculation methodology allows the consistent assessment of the hazard identified in the bow tie “risk of flashover to overline structure” (see module 3).

8.3.1.3 Guidance

Designs for new and reconstructed overline structures should provide functional (static) as a minimum where reasonably practical.

The clearance around an insulator (including a bridge arm) are sometimes tapered. Details of the tapered clearance for insulators are given in Appendix A.

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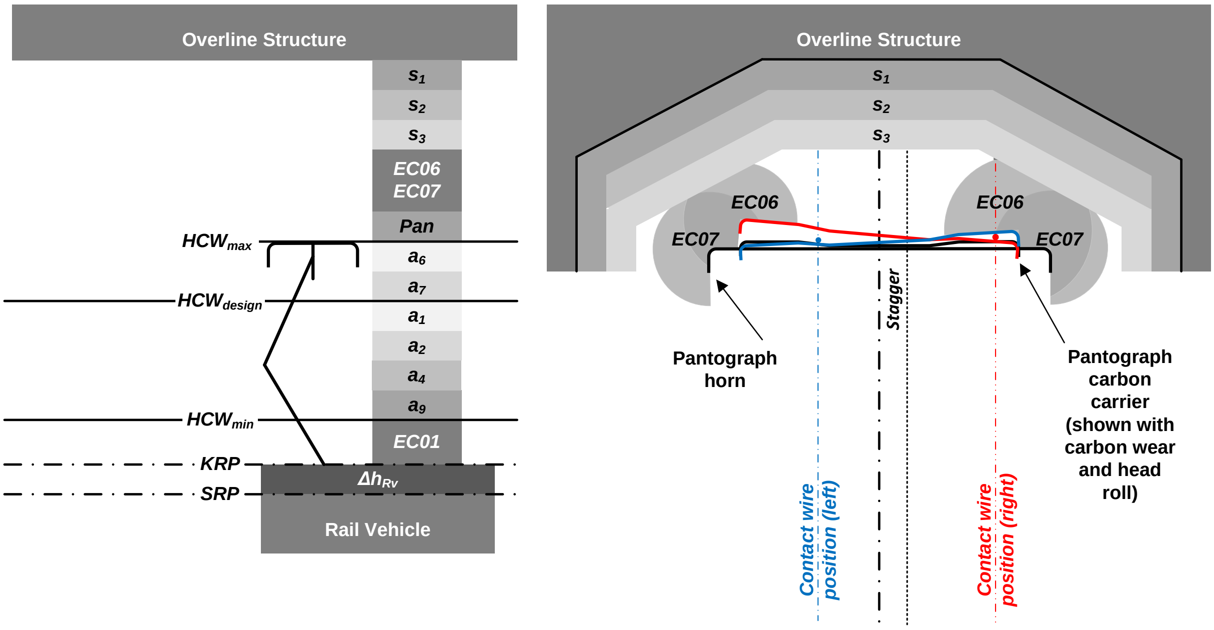
8.4 Pantographs to overline structures [EC06, EC07]

8.4.1.1 Requirements

The clearance between the pantograph carbon carrier [EC06] and horn [EC07] and overline structures shall be calculated in accordance with Figure 6.

No additional parameters or allowances shall be used for the MVP design option.



a_1	vertical upwards tolerance of the track
a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_6	uplift of the contact wire by the pantograph
a_7	upwards installation tolerance for the conductors
a_9	contact wire pre-sag
s_1	structure measurement tolerance
s_2	structure deflection
s_3	insulated coating
Pan	pantograph profile
Δh_{Rv}	vertical superelevation
stagger	contact wire stagger

Figure 6 – Electrical clearance diagram [EC06, EC07]

The insulation category of the minimum dimension for EC06 and EC07 shall be assessed using tables in module 5 and parameters from module 4 of this standard.

The pantograph head profile shall be calculated with clause 14 of this module. The profile to be used shall be either:

- a pantograph profile – active (probabilistic) for all arched overline structures where the mechanical clearance is less than 200 mm;

or

- a pantograph head profile – active in all other locations.

The designer shall prioritise achieving the highest electrical insulation category in accordance with Table 7 without significant cost and Table 24.

	Category				
	1	**2**	**3**	**4**	**5**
EC06	Reinforced	Basic	Functional (dynamic)	Basic (VCC)	Functional (stochastic)
EC07	Reinforced	Basic	Functional (dynamic)	Basic (VCC)	Functional (stochastic)

Table 7 – Electrical insulation category [EC06, EC07]

8.4.1.2 Rationale

This methodology aligns with BS EN 50119, BS EN 50367and GMRT2173.

This calculation methodology allows the consistent assessment of the hazards identified in the bow ties “risk of flashover to overline structure” and “risk of collision with overline structure” (see module 3).

8.4.1.3 Guidance

Composite pantograph horns have different electrical properties and therefore the minimum clearances for the pantograph horn and the pantograph carbon carrier need to be separately determined.

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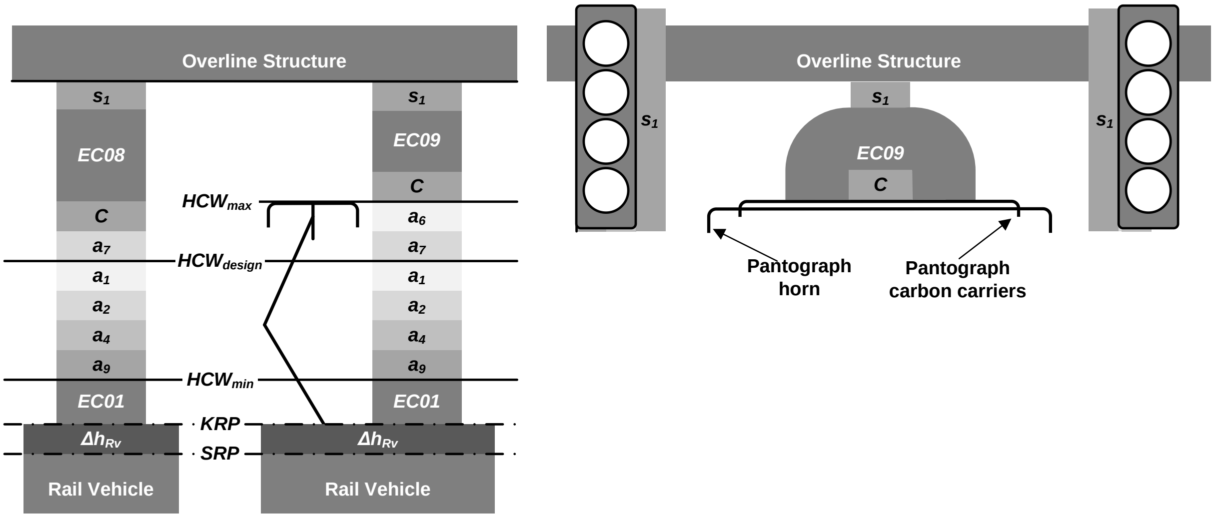
8.5 OCL to signals [EC08, EC09]

8.5.1.1 Requirements

The electrical clearance between the OCLS and signals shall be calculated in accordance with Figure 7.

Signals fitted with access platforms shall be additionally assessed using the restricted non-public standing surface requirements (see subclause 8.16).

No additional parameters or allowances shall be used for the MVP design option.



a_1	vertical upwards tolerance of the track
a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_6	uplift of the contact wire by the pantograph
a_7	upwards installation tolerance for the conductors
a_9	contact wire pre-sag
s_1	survey tolerance
C	conductors (contact wire and catenary, or double contact wire plus contact wire cover)
Δh_{Rv}	vertical superelevation

Figure 7 – Electrical clearance diagram [EC08, EC09]

The insulation category of the minimum dimension for EC08 and EC09 shall be assessed using tables in module 5 and parameters from module 4 of this standard.

The designer shall prioritise achieving the highest electrical insulation category in accordance with Table 8 and Table 24 without significant cost.Table 24

	Category				
	1	**2**	**3**	**4**	**5**
EC08	Reinforced	Basic	Functional (static)	Basic (VCC)	Functional (stochastic)
EC09	Reinforced	Basic	Functional (dynamic)	Basic (VCC)	Functional (stochastic)

Table 8 – Electrical insulation category [EC08, EC09]

8.5.1.2 Rationale

This methodology aligns with BS EN 50119 figure 3, BS EN 50367 subclause 5.2.2, GLRT1210 subclause 3.1.3.2 and GMRT2173 subclause 3.2.

This calculation methodology allows the consistent assessment of the hazards identified in the bow ties “risk of flashover to overline structure” and “risk of collision with overline structure” (see module 3).

Clearances to personnel accessing the signal platforms are assessed separately (see clause 15).

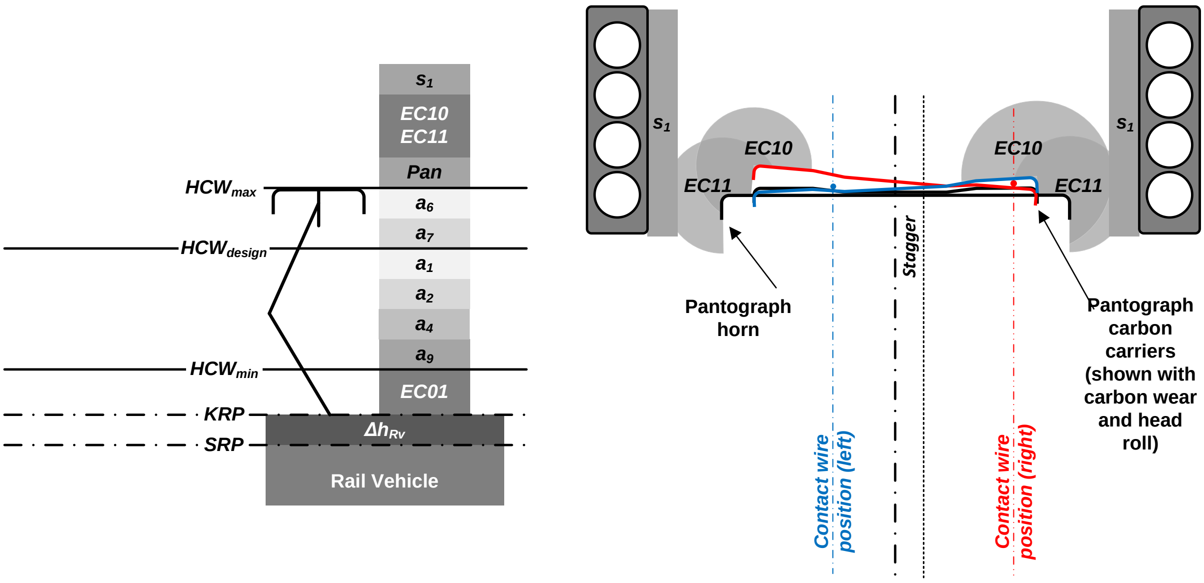
Ref:	NR/L2/ELP/27716/01
Issue:	1
Date:	02 December 2023
Compliance date:	02 March 2024

8.6 Pantographs to signals [EC10, EC11]

8.6.1.1 Requirements

The electrical clearance between the pantograph carbon carrier [EC10] and horn [EC11] and signals shall be calculated in accordance with Figure 8.

No additional parameters or allowances shall be used for the MVP design option.



a_1	vertical upwards tolerance of the track
a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_6	uplift of the contact wire by the pantograph
a_7	upwards installation tolerance for the conductors
a_9	contact wire pre-sag
s_1	survey tolerance
Pan	pantograph profile
Δh_{Rv}	vertical superelevation
stagger	contact wire stagger

Figure 8 – Electrical clearance diagram [EC10, EC11]

The insulation category of the minimum dimension for EC10 and EC11 shall be assessed using tables in module 5 and parameters from module 4 of this standard.

The pantograph head profile shall be calculated with clause 14 of this module. The profile to be used shall be either:

- a pantograph profile – active (probabilistic) for all overline signals where the mechanical clearance is less than 200 mm;

or

- a pantograph head profile – active in all other locations.

The designer shall prioritise achieving the highest electrical insulation category in accordance with Table 9 and Table 24 without significant cost.

	Category				
	1	**2**	**3**	**4**	**5**
EC10	Reinforced	Basic	Functional (dynamic)	Basic (VCC)	Functional (stochastic)
EC11	Reinforced	Basic	Functional (dynamic)	Basic (VCC)	Functional (stochastic)

Table 9 – Electrical insulation category [EC10, EC11]

8.6.1.2 Rationale

This methodology aligns with BS EN 50119 figure 3, BS EN 50367 subclause 5.2.2, GLRT1210 subclause 3.1.3.2 and GMRT2173 subclause 3.2.

This calculation methodology allows the consistent assessment of the hazards identified in the bow ties “risk of flashover to overline structure” and “risk of collision with overline structure” (see module 3).

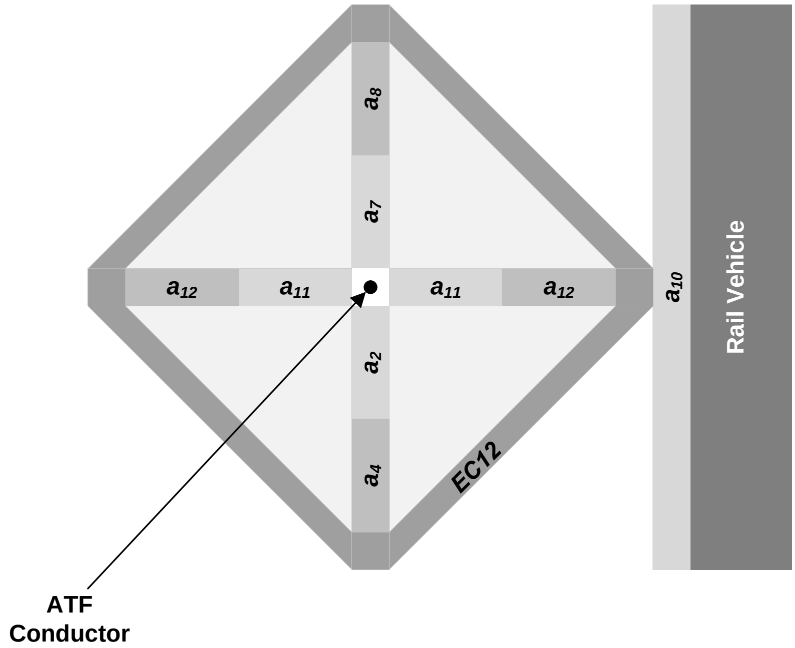
Clearances to personnel accessing the signal platforms are assessed separately (see clause 15).

Ref:	NR/L2/ELP/27716/01
Issue:	1
Date:	02 December 2023
Compliance date:	02 March 2024

8.7 ATF to rail vehicles bodies [EC12]

8.7.1.1 Requirements

The clearance between ATF and rail vehicles shall be calculated in accordance with Figure 9. The position of the rail vehicle shall be determined using a standard vehicle gauge.



a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_7	upwards installation tolerance for the conductors
a_8	upwards effects of conductor temperature
a_{10}	horizontal tolerance of track
a_{11}	horizontal tolerance for the conductors
a_{12}	effects of wind on lateral position of conductor

Figure 9 – Electrical clearance diagram [EC12]

The insulation category of the minimum dimension for EC12 shall be assessed using tables in module 5 and parameters from module 4 of this standard.

The designer shall prioritise achieving the highest electrical insulation category in accordance with Table 10 and Table 25 without significant cost.

	Category				
	1	**2**	**3**	**4**	**5**
EC12	Reinforced	Basic	Functional (dynamic)	Functional (stochastic)	–

Table 10 – Electrical insulation category [EC12]

8.7.1.2 Rationale

This calculation methodology allows the consistent assessment of the hazards identified in the bow tie “risk of flashover to rail vehicle” (see module 3).

8.7.1.3 Guidance

The ATF can be installed in a number of different arrangements to achieve electrical clearances. See clause 11.

The shape of EC12 in Figure 9 assumes tolerances and assumptions are not related. In reality, interdependency do exist (for example between the reduction in conductor sag due to cold temperatures and the effect of wind on the conductor). It is permitted to use these interdependencies to change the shape of EC12.

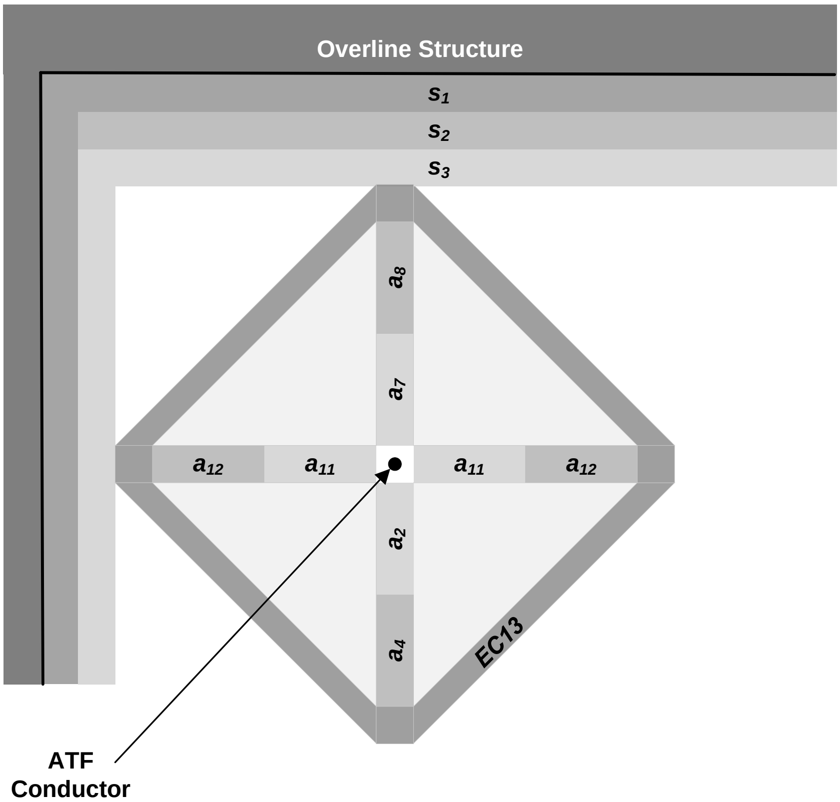
Ref:	**NR/L2/ELP/27716/01**
Issue:	**1**
Date:	02 December 2023
Compliance date:	02 March 2024

8.8 ATF to overline structures [EC13]

8.8.1.1 Requirements

The clearance between the ATF and overline structures shall be calculated accordance with Figure 10.

No additional parameters or allowances shall be used for the MVP design option.



a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_7	upwards installation tolerance for the conductors
a_8	upwards effects of conductor temperature
a_{11}	horizontal tolerance for the conductors
a_{12}	effects of wind on lateral position of conductor
s_1	structure measurement tolerance
s_2	structure deflection
s_3	insulated coating

Figure 10 – Electrical clearance diagram [EC13]

For all ATF apart from return conductors, the insulation category of the minimum dimension for EC13 shall be assessed using tables in module 5 and parameters from module 4 of this standard.

The designer shall achieve the electrical insulation category in accordance with Table 11 and Table 25 without significant cost.

	Category				
	1	**2**	**3**	**4**	**5**
EC13	Reinforced	Basic	Functional (static)	Functional (stochastic)	—

Table 11 – Electrical insulation category [EC13]

The minimum clearances for return conductors shall be:

- 50mm at a return conductor support;
- 600mm in span.

8.8.1.2 Rationale

This calculation methodology allows the consistent assessment of the hazard identified in the bow tie "risk of flashover to overline structure" (see module 3).

The return conductor clearance requirements form part of the railway reference system and are historic practice as described in NR/SP/ELP/21078 issue 2 and NR/L2/ELP/27009/MOD C49 issue 1.

8.8.1.3 Guidance

The ATF can be installed in a number of different arrangements to achieve electrical clearances. See clause 11.

The shape of EC13 in Figure 10 assumes tolerances and assumptions are not related. In reality, interdependencies exist (for example between the reduction in conductor sag due to cold temperatures and the effect of wind on the conductor). It is permitted to use these interdependencies to change the shape of EC13.

Ref:	NR/L2/ELP/27716/01
Issue:	1
Date:	02 December 2023
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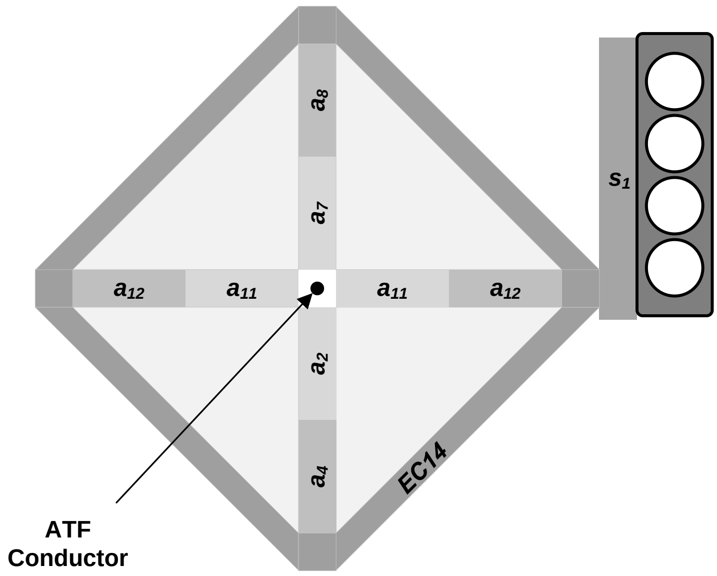
8.9 ATF to signals [EC14]

8.9.1.1 Requirements

The clearance between the ATF and signal structures shall be calculated in accordance with Figure 11.

No additional parameters or allowances shall be used for the MVP design option.



a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_7	upwards installation tolerance for the conductors
a_8	upwards effects of conductor temperature
a_{11}	horizontal tolerance for the conductors
a_{12}	effects of wind on lateral position of conductor
s_1	structure measurement tolerance

Figure 11 – Electrical clearance diagram [EC14]

The insulation category of the minimum dimension for EC14 shall be assessed using tables in module 5 and parameters from module 4 of this standard.

The designer shall achieve the electrical insulation category in accordance with Table 12 and Table 25 without significant cost.

	Category				
	1	**2**	**3**	**4**	**5**
EC14	Reinforced	Basic	Functional (static)	Functional (stochastic)	-

Table 12 – Electrical insulation category [EC14]

8.9.1.2 Rationale

This allows the consistent assessment of the following hazards:

- risk of flashover caused by proximity of overline structure;
- risk of flashover caused by low contact wire;
- risk of pantograph collision with overline structure caused by insufficient mechanical clearance;
- risk of electric shock to person caused by insufficient clearances.

See module 3 for more details.

8.9.1.3 Guidance

The ATF can be installed in a number of different arrangements to achieve electrical clearances. See clause 11.

The shape of EC14 in Figure 11 assumes tolerances and assumptions are not related. In reality, interdependencies exist (for example between the reduction in conductor sag due to cold temperatures and the effect of wind on the conductor). It is permitted to use these interdependencies to change the shape of EC14.

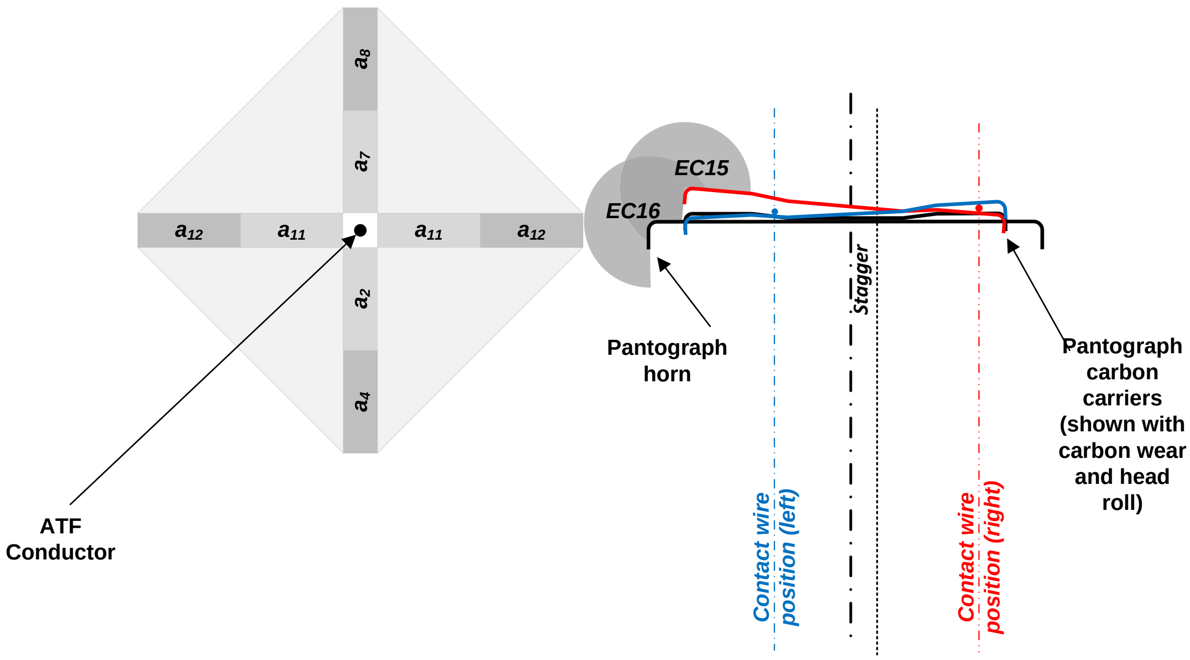
Clearances to personnel accessing the signal platforms are assessed separately (see clause 15).

8.10 ATF to pantographs [EC15, EC16]

8.10.1.1 Requirements

The clearance between the ATF and the pantograph carbon carriers [EC15] and horn [EC16] shall be calculated in accordance with Figure 12.

No additional parameters or allowances shall be used for the MVP design option.



a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_6	uplift of the contact wire by the pantograph (not shown)
a_7	upwards installation tolerance for the conductors
a_8	upwards effects of conductor temperature
a_9	contact wire pre-sag (not shown)
a_{11}	horizontal tolerance for the conductors
a_{12}	effects of wind on lateral position of conductor

Figure 12 – Electrical clearance diagram [EC15, EC16]

The insulation category of the minimum dimension for EC15 and EC16 shall be assessed using parameters from module 4 of this standard.

The pantograph head profile shall be calculated with clause 14 of this module. The profile to be used shall be pantograph head profile – active.

The designer shall prioritise achieving the highest electrical insulation category in accordance with Table 13 without significant cost.

	Category				
	1	**2**	**3**	**4**	**5**
EC15 120°	600 mm	400 mm	230 mm	–	–
EC15 180°	600 mm	540 mm	300 mm	–	–
EC16 120°	600 mm	400 mm	230 mm	–	–
EC16 180°	600 mm	540 mm	300 mm	–	–

Table 13 – Electrical insulation category [EC15, EC16]

8.10.1.2 Rationale

This methodology aligns with BS EN 50119, BS EN 50367, and GMRT2173 and mitigates the risk of phase-to-phase flashover.

8.10.1.3 Guidance

For 25 kV autotransformer systems, there is a phase difference of 180° between all live parts connected to the feeder line and all live parts connected to the OCLS.

Ref:	NR/L2/ELP/27716/01
Issue:	1
Date:	02 December 2023
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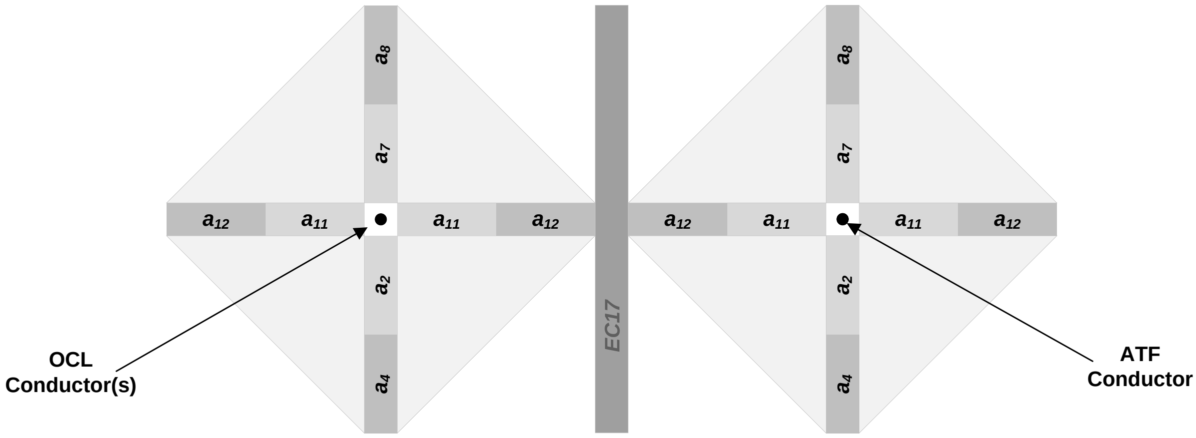
8.11 OCL conductors to ATF [EC17]

8.11.1.1 Requirements

The clearance between OCL electrical sections and other energised conductors (autotransformer feeder, return conductor etc.) shall be calculated in accordance with Figure 13.

No additional parameters or allowances shall be used for the MVP design option.



a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_7	upwards installation tolerance for the conductors
a_8	upwards effects of conductor temperature
a_9	contact wire pre-sag (not shown)
a_{11}	horizontal tolerance for the conductors
a_{12}	effects of wind on lateral position of conductor

Figure 13 – Electrical clearance diagram [EC17]

The insulation category of the minimum dimension for EC17 shall be assessed using tables in module 5 and parameters from module 4 of this standard.

The designer shall prioritise achieving the highest electrical insulation category in accordance with Table 14 the phase difference.

	Category				
	1	**2**	**3**	**4**	**5**
EC17 (120°)	600mm	400 mm	-	-	-
EC17 (180°)	600mm	540 mm	-	-	-

Table 14 – Acceptable electrical category [EC17]

8.11.1.2 Rationale

This methodology aligns with BS EN 50119 and mitigates the risk of phase-to-phase flashover.

8.11.1.3 Guidance

It is recognised that it is improbable for the wind pressure to create opposing movement in the conductors simultaneously. It is therefore acceptable to calculate the clearances using different wind pressure load cases.

Autotransformer feeders and return conductors are energised with a 180° phase difference.

Ref:	NR/L2/ELP/27716/01
Issue:	1
Date:	02 December 2023
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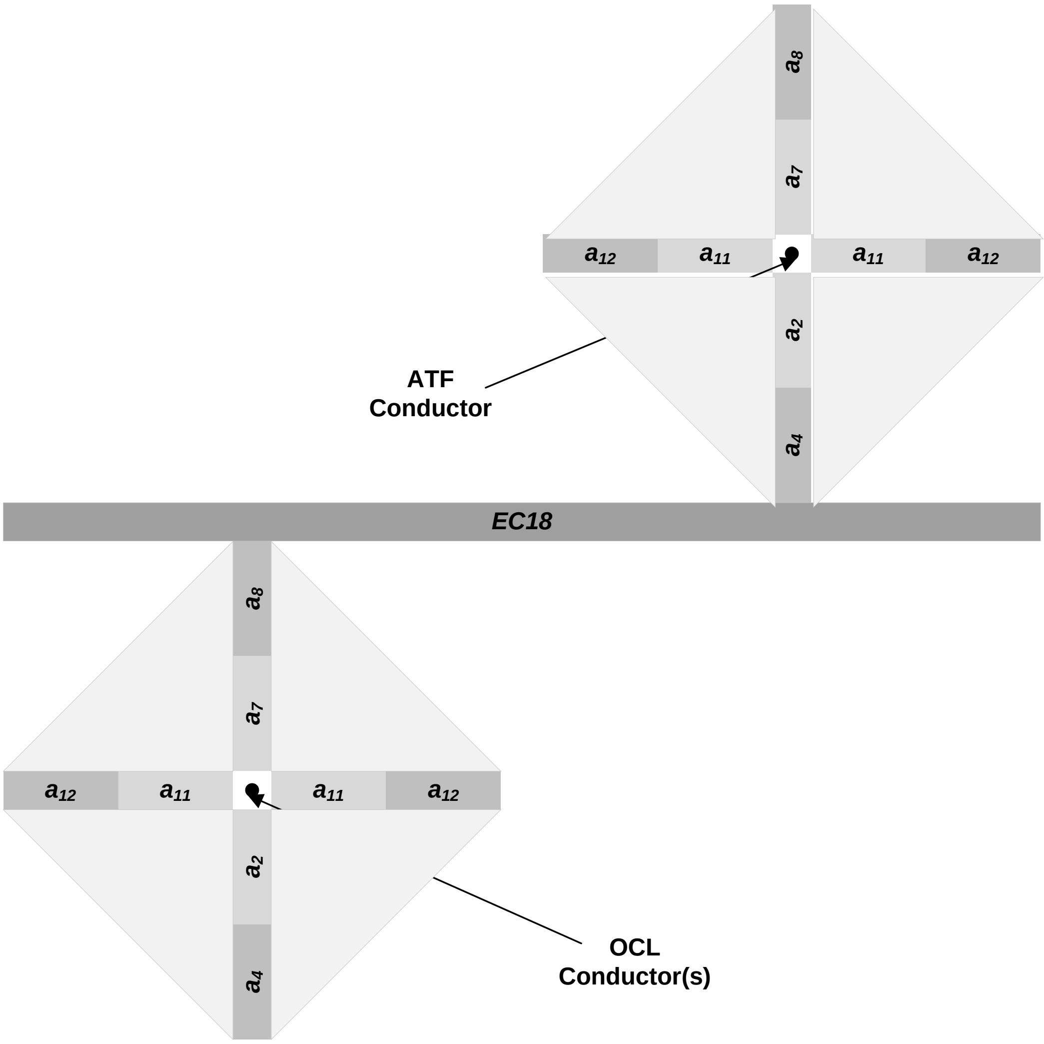
8.12 OCL conductors to ATF – vertical working clearance [EC18]

8.12.1.1 Requirements

The vertical clearance between OCL electrical sections or other energised conductors (ATF, cross track feeders etc.) shall be calculated in accordance with Figure 14.

No additional parameters or allowances shall be used for the MVP design option.



a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_7	upwards installation tolerance for the conductors
a_8	upwards effects of conductor temperature
a_{11}	horizontal tolerance for the conductors
a_{12}	effects of wind on lateral position of conductor

Figure 14 – Electrical clearance diagram [EC18]

The route operations and maintenance strategy shall detail the planned maintenance arrangements for the route. This document shall also detail the planned energisation state of each of the ATF and OCL for maintenance activities.

The designer shall provide a minimum vertical clearance to support the maintenance strategy agreed with the route. The minimum vertical clearance shall be aligned with Table 15.

Option	OCL	ATF	Working limit	Minimum vertical clearance between OCL and ATF
1	Dead	Dead	None	No additional requirement
2	Dead	Charged	Below contact wire	600 mm
3	Dead	Live	OCL only	1600 mm

Table 15 – Vertical safety clearances

8.12.1.2 Rationale

This methodology meets requirements in:

- NR/L2/ELP/25001 – Electrical safety principles for new electrification
- NR/L3/ELP/29987 – Working on or about electrified lines.

8.12.1.3 Guidance

A risk assessment is required by NR/L3/ELP/29987 for working closer than 2.75 m to live equipment.

This does not apply to aerial earth wires (AEWs).

8.13 OCLS conductors to other OCLS conductors [EC19]

8.13.1.1 Requirement

The clearance (EC19) between different OCLS electrical sections or part sections (i.e. insulated overlap) shall achieve the values in Table 16.

Clearance ID	Static				
EC19	460 mm	–	–	–	–

Table 16 – Electrical clearance [EC19]

8.13.1.2 Rationale

This is a historical value used for insulated overlaps.

8.13.1.3 Guidance

Further guidance on insulated overlaps is provided in the OCLS design ranges.

Section insulator requirements are detailed in NR/L2/ELP/27428 Module 2. These include electrical impulse testing to BS EN 50124-1:2017.

Ref:	**NR/L2/ELP/27716/01**
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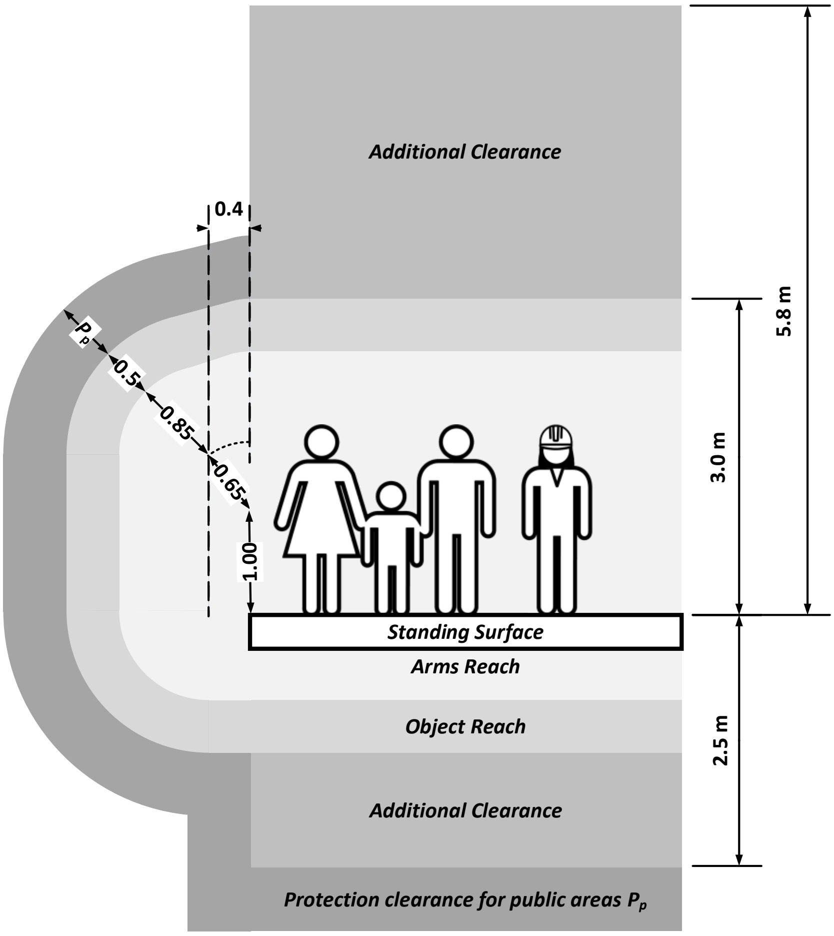
8.14 OCLS and pantographs to standing surfaces (non-restricted public areas) [EC20, EC21, EC22]

8.14.1.1 Requirements

The position of live conductors including OCLS (EC20), the pantograph carbon carrier (EC21) and horn (EC22) shall be assessed using Figure 15.



	legs	1 m
	torso	0.65 m
	arm	0.85 m
	object (hand-held)	0.5 m
P_p	protective clearance in public areas	0.6 m for 25 kV

Figure 15 – OCLS to non-restricted public areas [EC20, EC21, EC22]

The parameters used to determine the position of live conductors shall be based on the following:

- OCLS as per clause 10;
- ATF conductors as per clause 11;
- Pantograph head profile – static for stationary rail vehicles as per clause 14.

No live OCLS conductors shall enter the protective clearance or additional clearance zones.

If the pantograph head profile – static protrudes into the P_p zone, then following shall be considered:

1. move train stop position so the pantographs does not stop at low contact wire position; or
2. increase the gradient of the contact wire between the overline structure and the pantograph stop position; or
3. if neither of these are reasonably practical, the designer shall follow the hierarchy listed in Table 17.

No.	**Scenario**	**Protective clearances** P_p – **Pantograph carbon carrier**	**Protective clearances** P_p - **Pantograph horn**	**Control measures CM10**	**Control measures CM11**	**Control measures CM12**
1	BS EN 50122-1:2022 Object + *Pp*	500 + 600 = 1100 mm	500 + 600 = 1100 mm	No		
2	Reinforced insulation – object + conductive pantograph horn	500 + 450 = 950 mm	500 + 450 = 950 mm	No		
3	Reinforced insulation – object + insulated pantograph horn	500 + 450 = 950 mm	500 + 200 = 700 mm	Yes		
4	Reinforced insulation – 50% object + conductive pantograph horn	250 + 450 = 700 mm	250 + 450 = 650 mm	No	Yes	
5	Reinforced insulation – 50% object + insulated pantograph horn	250 + 450 = 700 mm	250 + 200 = 450 mm	Yes	Yes	
6	Reinforced insulation – no object + conductive pantograph horn	0 + 450 = 450 mm	0 + 450 = 450 mm	No	Yes	Yes
7	Reinforced insulation – no object + insulated pantograph horn	0 + 450 = 450 mm	0 + 200 = 200 mm	Yes	Yes	Yes

CM10 – Stopping trains fitted with insulated pantograph horns

CM11 – Warning signs installed on platforms – NR/L2/ELP/21131 RTE 6018

CM12 – Site specific risk assessment required

Table 17 – Pantograph clearance hierarchy for non-restricted public areas

An update to the RIS-3703-TOM platform risk assessment shall be produced by the designer, and provided to the railway undertaking / infrastructure manager responsible for the station platform.

8.14.1.2 Rationale

The requirement is based on the following documents:

- GLRT1210 issue 3 subclause 3.3 stipulates the use of BS EN 50122-1:2022 subclauses 5.1, 5.2 & 5.3.
- BS EN 50122-1:2022 figure 5 provides the minimum distance to accessible live parts from a standing surface with figure 4 detailing the manikin dimensions.
- BS EN 50122-1:2022 subclause 5.1.3 permits the substitution of the protection clearance (P_p) with a clearance derived from BS EN 50124:2017. The values in Table 17 have been derived by testing independently for both conductive and insulated pantograph horns.
- BS EN 50122-1:2022 subclause 5.2.2 permits the use of the deviating dimension (i.e. the reduction of object length by 50%) detailed in subclause 5.4.
- RIS-3703-TOM details the requirement for every station platform to have a risk assessment.
- ORR's policy on electrical clearances to standing surfaces for 25 kV overhead electrification.

ORR policy states:

> *while we recognise that they [transiting pantographs] pose an electrical risk, the presence of a rail vehicle moving at speed is likely to present a greater risk. So provided suitable controls are in place to ensure that the risk from being struck by a moving vehicle is managed then we believe that electrical risk from passing pantographs can be considered negligible.*

8.14.1.3 Guidance

Non-restricted public areas are generally station platforms. Further details are contained in module 3.

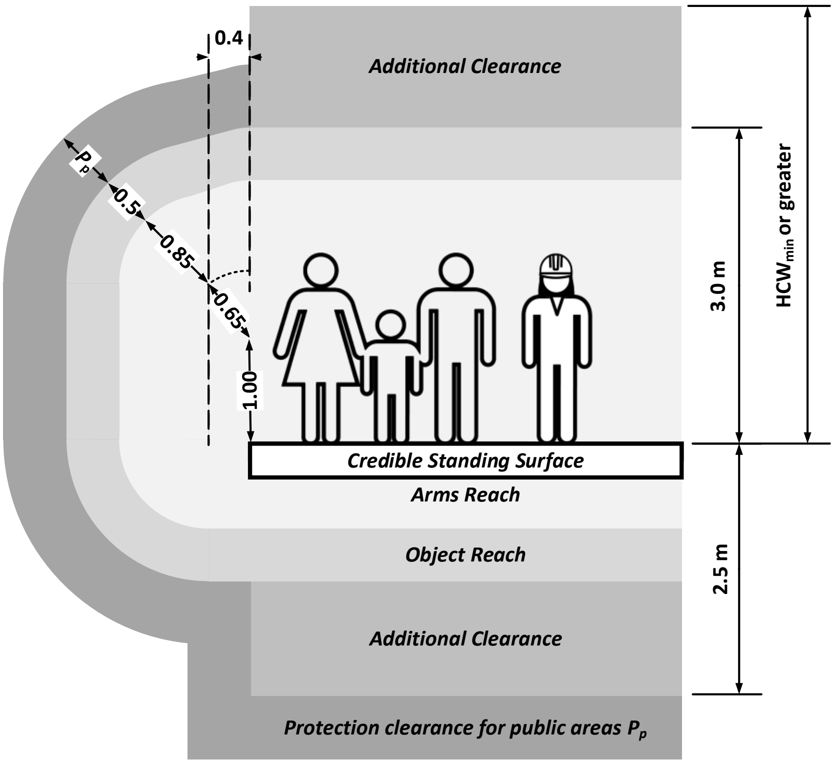
Ref:	**NR/L2/ELP/27716/01**
Issue:	**1**
Date:	02 December 2023
Compliance date:	02 March 2024

8.15 OCLS and pantographs to standing surfaces (restricted lawful public areas) [EC23, EC24, EC25]

8.15.1.1 Requirements

The position of live conductors including OCLS (EC24), the pantograph carbon carrier (EC25) and horn (EC26) shall be assessed using Figure 16.

No live conductors shall enter the protective clearance or additional clearance zones.



	legs	1 m
	torso	0.65 m
	arm	0.85 m
	object (hand-held)	0.5 m
P_p	protective clearance in public areas	0.6 m for 25 kV

Figure 16 – OCLS to restricted public areas [EC23, EC24, EC25]

The parameters used to determine the clearances shall be based on the following:

- OCLS as per clause 10,
- ATF conductors as per clause 11,
- Pantograph head profile – static for stationary rail vehicles as per clause 14.

If the pantograph head profile – static protrudes into the P_p zone, then the considerations, hierarchy and control measures described in subclause 8.14.1.1 shall be applied.

8.15.1.2 Rationale

The requirement is based on the following documents:

- BS EN 50122-1:2022 figure 5
- ORR's policy on electrical clearances to standing surfaces for 25 kV overhead electrification.

8.15.1.3 Guidance

Restricted lawful public areas include only those areas where passengers may be de-trained and escorted by railway staff. These locations are limited to the lineside cess, authorised walking routes and authorised access points. For example, public are not lawfully permitted to stand on working platforms, cutting slopes, retaining walls or bridge wingwalls during a de-training event and these locations are not considered as restricted lawful public areas. Those locations should be assessed in accordance with subclause 8.16.

A stationary rail vehicle usually only interfaces with restricted lawful public areas when a train is stopped behind a signal.

This requirement is not mandated by NTSN or GLRT1210, and therefore falls outside the scope of assessments conducted by UK approved and designated bodies.

8.16 OCLS and pantographs to standing surfaces (restricted non-public areas) [EC26, EC27, EC28]

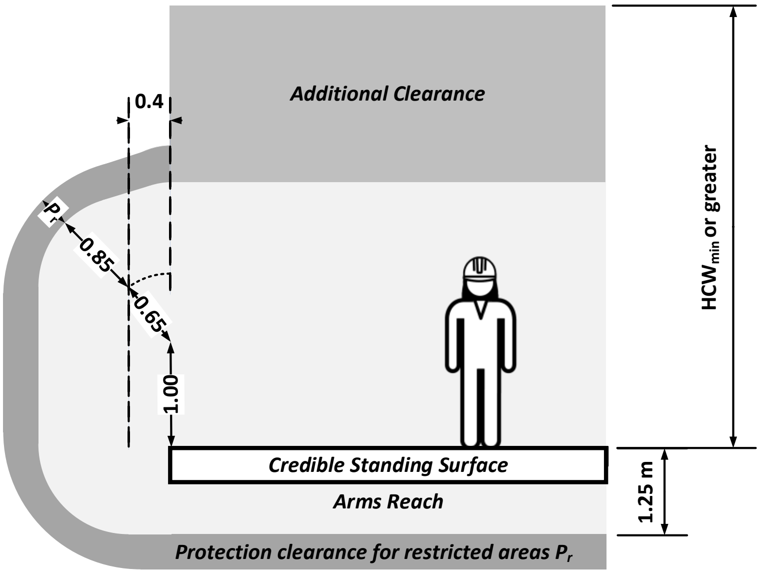
8.16.1.1 Requirements

The position of live conductors including OCLS (EC26), the pantograph carbon carrier (EC27) and horn (EC28) shall be assessed using Figure 17.

No live conductors shall enter the protective clearance or additional clearance zones.

The parameters used to determine the clearances shall be based on the following:

- OCLS as per clause 10,
- ATF conductors as per clause 11,
- Pantograph head profile – static for stationary rail vehicles as per clause 14.



	legs	1 m
	torso	0.65 m
	arm	0.85 m
P_r	protective clearance in restricted areas	0.3 m for 25 kV

Figure 17 – OCLS to restricted non-public areas [EC26, EC27 and EC28]

A static pantograph profile shall be used when assessing clearances to a credible standing surface.

8.16.1.2 Rationale

The requirement is based on the following documents:

- BS EN 50122-1:2022 figure 5, with a protection clearance (P_r) of 0.3 m,
- ORR's policy on electrical clearances to standing surfaces for 25 kV overhead electrification.

8.16.1.3 Guidance

Restricted non-public areas include working platforms for signals and location cases, and station roofs.

In line with typical risk assessment practices, the approach should be to eliminate the hazard wherever practicable. Therefore the removal of the standing surface would remove the need for additional works. Examples given for guidance are:

- closing any standing surfaces not in use, e.g. platforms;
- removing sections of the standing surface that are non-compliant;
- installation of obstacles (screening etc.);
- adjusting the height of the standing surface.

In many cases it is envisaged that removal will not be a practicable option, in which case the designer should provide justification for this before commencing the OCLS design.

8.17 OCLS to level crossings and other highways crossings [EC29]

8.17.1.1 Requirements

At road vehicle level crossing, footpath crossings and other highways crossings (including private roads and farm tracks), the designer shall design the OCLS to achieve the minimum heights detailed in Table 18.

The road surface shall be determined as the design road level directly below the live equipment.

As part of design development, the relevant highways agency shall be consulted regarding any design proposal.

No.	HCW_{min}	Location	Additional requirements
1	5.8 m	Vehicle crossings	–
2	5.6 m + S	Vehicle crossings	Vertical road curvature assessment required
3	5.2 m	Bridleway crossings	–
4	HCW_{min}	Footpath crossings	–

Table 18 – Minimum conductor heights at level crossings

When option 2 is selected from Table 18 then an additional clearance S from Table 19 shall be applied to the HCW_{min}.

Vertical road curvature (concave) (m)	Additional clearance S (mm)
≤ 1000	80
1200	70
1500	55
200	45
3000	25
6000	15
> 6000	0

Table 19 – Road curvature compensation (based on Design Manual for Road and Bridges, CD 127, table 4.3)

Bridleways shall include signage advising horse riders to dismount before crossing.

In Scotland, the Land Reform (Scotland) Act 2003 shall be taken into account when determining if a crossing can be classified as a footpath only.

8.17.1.2 Rationale

These requirements are based on the following documents:

- GLRT1210,
- The Electricity Safety, Quality and Continuity Regulations,
- Design Manual for Roads and Bridges,
- The Highway Code,
- ORR's policy on electrical clearances to standing surfaces for 25 kV overhead electrification,
- The Land Reform (Scotland) Act 2003,
- The DfT Traffic Signs Manual.

A detailed justification is provided in module 3 of this manual.

8.17.1.3 Guidance

Route level crossing managers maintain a separate risk assessment for each level crossing; the electrical clearances assessment should be done in coordination with them or appended to the existing risk assessment. If possible, the level crossing should be removed prior to electrification.

If the minimum height is not achievable due to other constraints, then a risk assessment may allow a lower height.

Section 178 of the Highways Act 1980 requires any person who places cables or wires or similar apparatus over, along or across a highway to first obtain the consent of the highway authority.

However, where these works are to be undertaken by a statutory undertaker, Section 178 (5) provides an exception. The definition of 'statutory undertaker' for the purpose of this section of the Highways Act 1980 is contained in section 329. Railways are considered a statutory undertaker, and this applies to Network Rail Infrastructure. Therefore the legal obligation is to consult only.

The Land Reform (Scotland) Act 2003 provides a right of access to land if exercised responsibly. Railways are considered a statutory undertaking by this legislation. This legislation provides rights of way by foot, horseback, pedal cycle or any combination of those providing these are responsibly used.

In accordance with NR/L2/ELP/24013, notification of energisation should be sent to bus operators in connection with open top buses.

8.18 OCLS to third party networks – electricity [EC30]

8.18.1.1 Requirements

The clearances between OCLS fixed equipment and third-party electricity networks shall achieve the vertical distances shown in Figure 18 and detailed in Table 20.

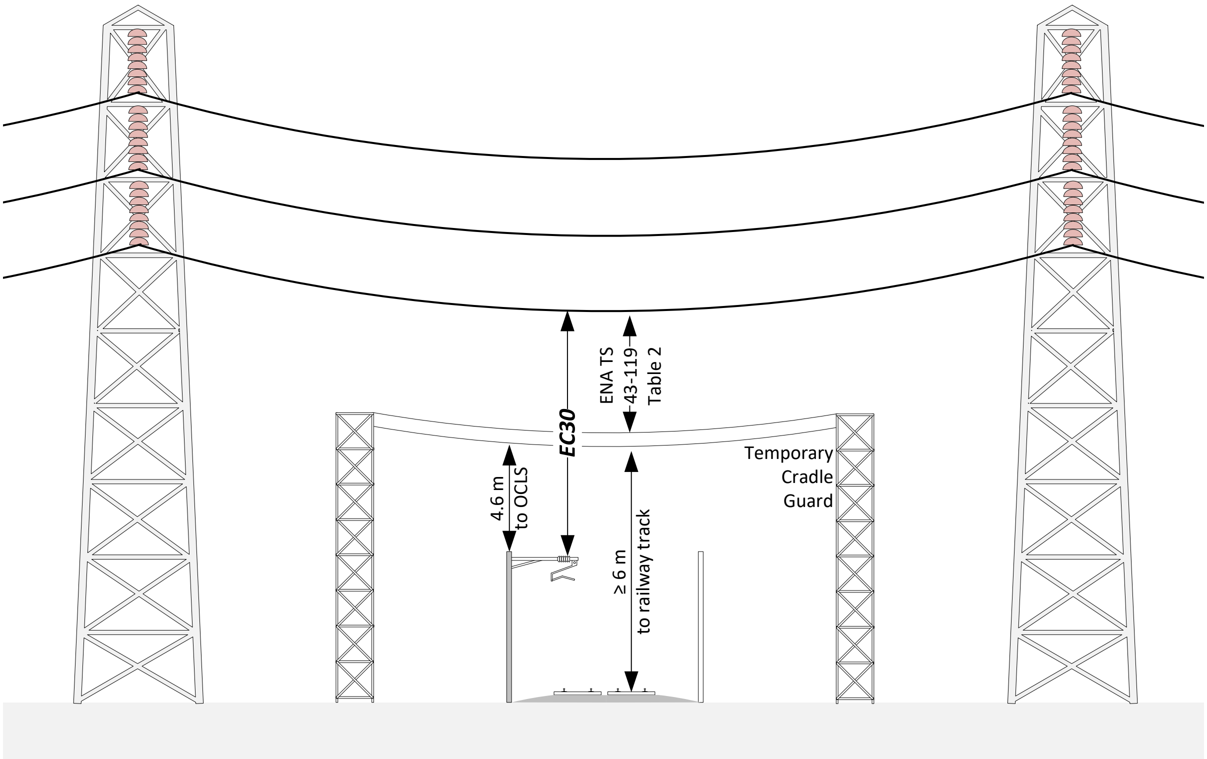


Figure 18 – Electrical clearance diagram [EC30]

	Nominal voltage of third-party overhead line				
	≤ 33 kV	66 kV	132 kV	275 kV	400 kV
Clearance EC30 from OCLS to electricity network conductors [m]	3.0	3.0	3.7	4.6	6.1

Table 20 – Principle vertical clearances to electricity network lines [EC30]

If a temporary cradle guard is required to support works on the electricity network, then an additional clearance of 4.6 m shall be provided between the cradle guard and the 25 kV live equipment. Further separation shall be provided between the cradle guard and the third-party network as specified by ENA Technical Specification 43-119.

To support the MVP approach, the contact wire and catenary shall be lowered to avoid modification to the electricity network whenever possible.

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8.18.1.2 Rationale

These requirements are based on:

- ENA Technical Specification (TS) 43-8 (table 6, row 2); and
- ENA TS 43-119.

8.18.1.3 Guidance

HSE GS6 references a booklet entitled "Look Out, Look Up" and this booklet advises establishing an exclusion zone of 6.0 m when working underneath DNO lines. "Look Out, Look Up" is published by ENA and provides a simplified message aimed at farmers and construction workers.

The booklet states "it is for guidance only, and if you are in any doubt, please call your local electricity company for advice before setting up your plant or starting work". Advice to the electricity companies is documented in ENA TS 43-8.

The dimensions in Table 20 for clearances from 25 kV equipment are taken from ENA TS 43-8 issue 5, subclause 11.4. This states "*The intent of a vertical clearance when working beneath a line is to ensure that the safety distance (see Table 8 item 1 and Table A.1) is maintained as a minimum at all times i.e. prevention of a person breaching the safety distance (see A.1). It is important that an appropriate safe system of work is employed and overseen by experienced and competent persons. The use of an 'application factor' is a relevant consideration, as described in A.2: an allowance of 2.2 m for a person to move their arm whilst holding a short metallic object."* This is an adaption of a rule based on imperial units so there are some rounding errors.

ENA TS 43-8 table 8 is repeated in Table 21 below.

	Nominal voltage of third-party overhead line				
	≤ 33 kV	66 kV	132 kV	275 kV	400 kV
Minimum vertical passing clearances [m]	0.8	1.0	1.4	2.4	3.1

Table 21 – Minimum vertical passing clearances to electricity network lines

Depending on the wayleave agreement, it may be possible not to provide the space for a cradle guard and to agree either:

- a future operation and maintenance strategy that requires a railway isolation to enable electricity network renewals,

or

- the future provision of cradle guard space (to align with electricity grid renewals), rather than provision of the space prior to the energisation of the 25 kV equipment.

The clearance to a permanent cable guard is an open point.

8.19 OCLS to third party networks – telecoms [EC31]

8.19.1.1 Requirement

The horizontal clearances between OCLS fixed equipment and third-party telecommunication networks shall achieve the minimum clearances in ENA TS 43-8.

Third-party telecommunication networks crossing an electrified railway shall be either:

- diverted into an under-track crossing, or away from the railway,

or

- installed in overline structures to avoid direct contact.

8.19.1.2 Rationale

These requirements are based on:

- ENA TS 43-8 issue 5, and
- ENA TS PO5.

8.19.1.3 Guidance

There is no legal requirement to comply with the dimensions within BT Openreach publication EPT/PPS/B023. However where this is possible and not grossly-disproportionately impact in both cost and programme, it should be considered by the designer.

8.20 OCLS to third party land [EC32]

8.20.1.1 Requirements

The clearances between credible standing surfaces on third party land (apart from the relevant highway authority) and OCLS (EC32) shall be assessed using Figure 19.

If an obstacle (e.g. a wall or fence) is present at, or close to, the boundary, then the obstacle shall be assessed in accordance with BS EN 50122-1:2022. This includes any openings (e.g. windows) in the obstacle.

The highway authority shall be consulted on any placement of conductors over a highway (including level crossings) in accordance with the Highways Act 1980.

Warning signs shall be installed on supporting OLE structures facing the boundary where 25 kV bare conductors are positioned within 2.75 m of the boundary.

8.20.1.2 Rationale

Figure 19 has been created using Figure 15 and Figure 16.

8.20.1.3 Guidance

Additional signage should be considered on the boundary fence.

The AEW is not considered a bare 25 kV conductor.

Land ownership in the UK also includes ownership of the airspace above that land.

For locations where the third-party land is a highway see subclause 8.17.1.3.

Under the Construction (Design and Management) Regulations 2015, it is the responsibility of a designer to identify and mitigate risk.

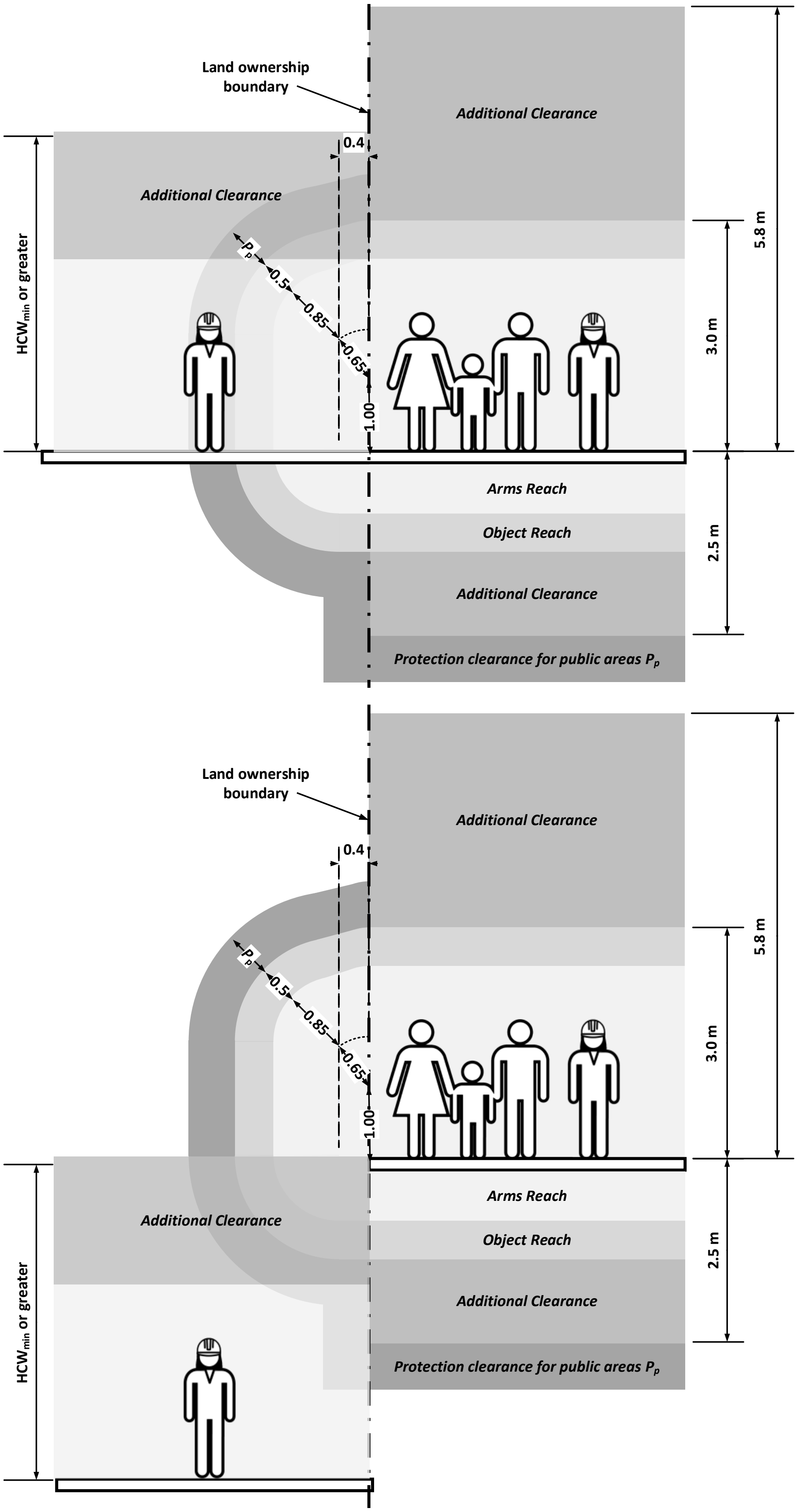

Figure 19 – Electrical clearance diagrams [EC32]

8.21 OCLS to vegetation [EC33]

8.21.1.1 Requirement

The clearance between the live fixed equipment and branches of trees or bushes shall comply with NR/L2/OTK/5201.

8.21.1.2 Rationale

None.

8.21.1.3 Guidance

Where possible the designer should place conductors to minimise the need for de-vegetation works both in the construction phase and for ongoing maintenance.

Specific attention should be given to locations of existing vegetation and where vegetation could be expected to grow in the near future.

The designer should not assume the possibility of removal of third party vegetation.

8.22 Single phase HV cables [EC35, EC36, EC37, EC38]

8.22.1.1 Requirements

Designers shall position single phase HV cables in a way that provides a separation distance between persons and excessive electromagnetic fields (EMF).

A maximum magnetic field strength of 360 μT ('field actually required') shall be used to determine that there is sufficient separation distance between persons and HV cables.

8.22.1.2 Rationale

The approach is consistent with the approach taken by:

- ENA SHE Standard 10 'EMFs – Occupational Exposure Limits', 2016 Edition, and
- The ENA–National Grid–Energy UK Risk Assessment 'Compliance of the UK Transmission and Distribution Networks and Generating Equipment with Occupational Exposure Limits', Issue 2.4.

8.22.1.3 Guidance

Electric and magnetic fields are produced wherever there are currents or voltages present. A larger current or voltage results in larger EMFs, and closer approach to the source likewise results in larger EMFs.

The limits of exposure and therefore separation distances are either based on the electric field (voltage magnitude dependent) or magnetic field (current magnitude dependent).

The electric field is difficult to calculate and generally requires computer modelling or measurement. The approach taken by Network Rail is to use the separation distances provided in RSSB supporting guidance note GLGN1620. For bare conductors the minimum approach distance to prevent electric shock is in excess of any electric and/or magnetic field limits.

The separation distance for magnetic fields can be calculated using the Biot Savart equation (see equation below and Table 22).

$$\beta = \frac{\mu_o I}{2\pi r}$$

β	magnetic field	(Tesla)
μ_0	permeability of free space	1.2566×10^{-06} N·A^{-2}
I	current	(Amps)
π	Pi	3.1416
r	separation distance	(metres)

Table 22 – Biot Savart equation inputs

The Biot Savart methodology assumes a single conductor and does not take into account a reduction in magnetic field as a result of the magnetic field from other conductors having a ‘cancelling effect’. The results from the Biot Savart calculation, from a single cable are shown in Figure 20.

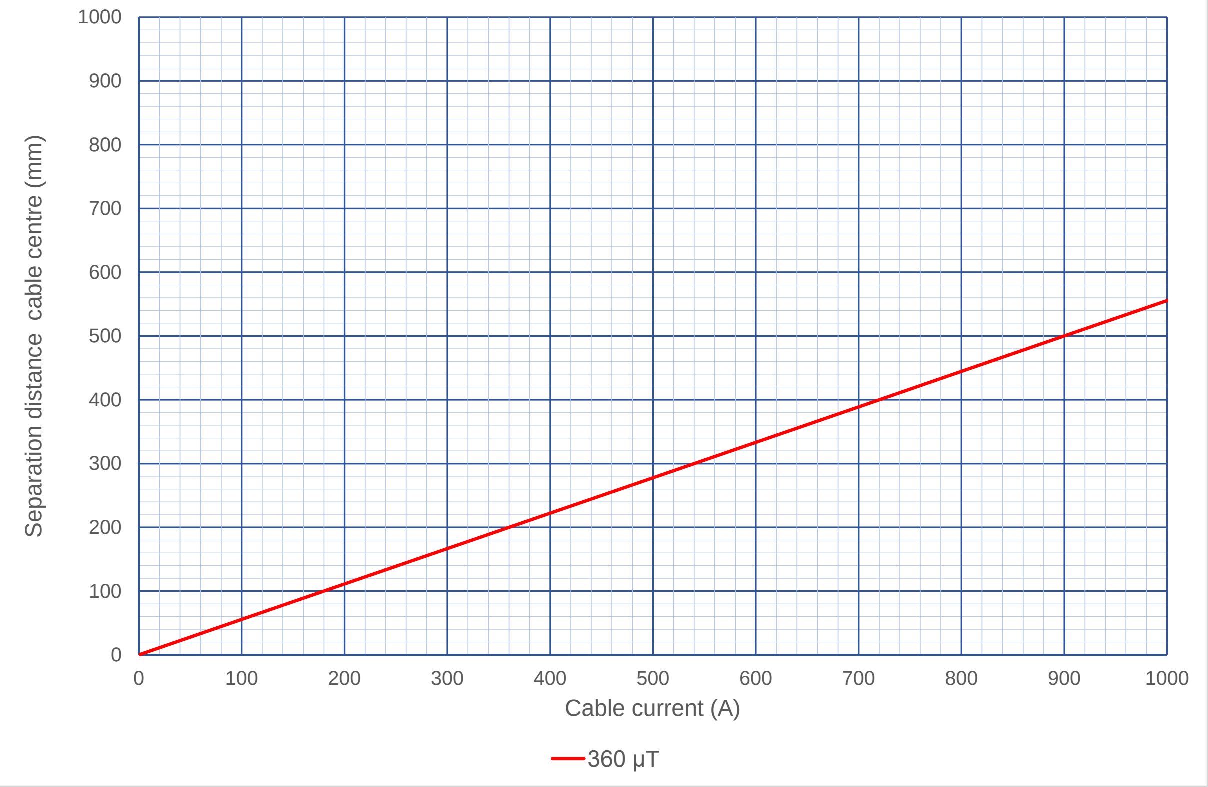

Figure 20 – Calculated separation distance from cable centre using the Biot Savart equation

In accordance with GLGN1620 issue 1, subclause 3.4.7, fault current levels are not included in the assessment.

Normal feeding should be assumed when determining the current loading. N−1 is unlikely to occur, so can be excluded from the EMF analysis. It is recommended that the continuous rating of the cable is used. Alternatively, the 15-minute average could be used.

Previous studies have identified the worst case to be elevated cable route and a track worker leaning against the cable route. In this case the distance between heart and ATF was less than 250mm.

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9 Mechanical clearances

9.1 Pantograph to overline structure [MC01, MC02, MC03, MC04, MC05]

9.1.1.1 Requirements

Clearance from the pantograph head shall be categorised in accordance with Table 23.

	Mechanical clearance category	Mechanical clearance	
		Minimum	**Maximum**
1	Normal clearance	≥ 100 mm	and above
2	Reduced clearance	≥ 50 mm	< 100 mm
3	Special reduced clearance	> 0 mm	< 50 mm

Table 23 – Mechanical clearance classification for line speeds up to 125 mph

Note: NR/L2/TRK/2102 requires normal clearance to be provided for vehicle gauges. This does not apply to the pantograph as the pantograph is not included in the vehicle gauge.

Mechanical clearances of 25 mm or less shall not be permitted without written Regional Engineer agreement.

9.1.1.2 Rationale

The requirement has been adapted from the gauge clearance requirements specified in GIRT7073.

9.1.1.3 Guidance

More information is contained within GIRT7073.

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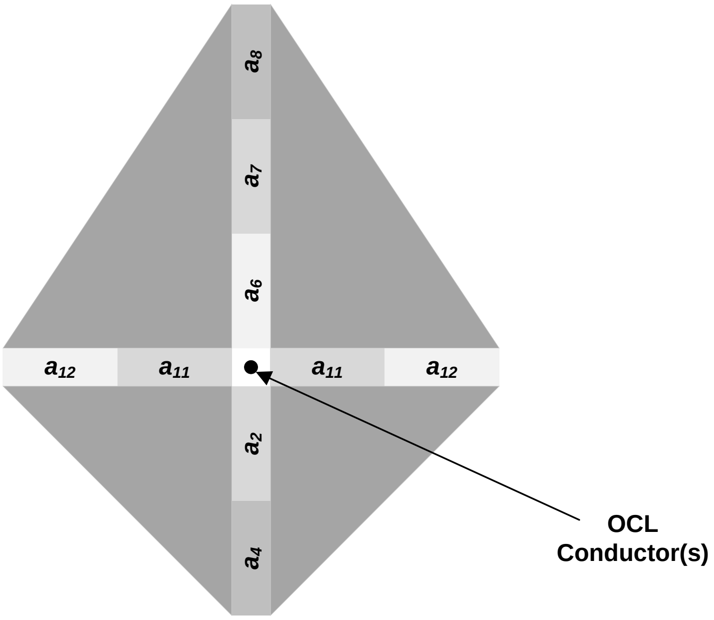
10 OCL design

10.1.1.1 Requirements

When determining if the OCL achieves a distance from a standing surface, then the OCL position shall be determined in accordance with Figure 21.

No additional parameters or allowances shall be used for the MVP design option.



a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_6	uplift of the contact wire by the pantograph
a_7	upwards installation tolerance for the conductors
a_8	upwards effects of conductor temperature
a_{11}	horizontal tolerance for the conductors
a_{12}	effects of wind on lateral position of conductor

Figure 21 – OCL design

When determining an appropriate design arrangement for overline structures (including signal structures), the hierarchy in Table 24 shall be followed.

Number	Insulation categories				OCLS arrangement				Contact wire height	
	Reinforced	Basic	Functional	Basic (VCC)	Free running	Reduced system depth	Twin contact	Equipment fitted	> 4165 mm	≤ 4165 mm
1	✓				✓				✓	
2		✓			✓				✓	
3	✓					✓		✓	✓	
4			✓		✓				✓	
5	✓						✓	✓	✓	
6		✓				✓		✓	✓	
7		✓					✓	✓	✓	
8			✓				✓	✓	✓	
9				✓			✓	✓	✓	✓

Table 24 – OCLS design arrangement hierarchy

The designer shall record the justification in the design decision log for each higher numbered preference (lower hierarchy) option not achieved before moving to the next option down (1 = most preferred) – see appendix C.

10.1.1.2 Rationale

This hierarchy was determined from analysis in module 3.

10.1.1.3 Guidance

The effects of temperature on OCL conductors only impacts fixed terminated (FT) equipment. For auto tensioned equipment the effect of temperature can be ignored.

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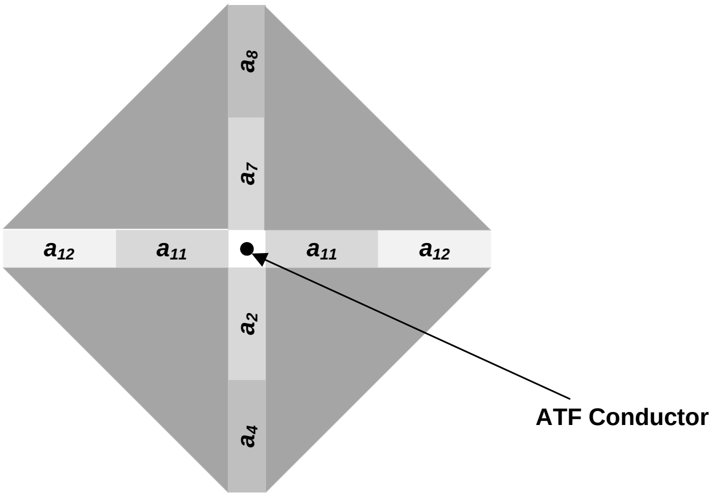
11 ATF design

11.1.1.1 Requirements

When determining if the ATF achieves a distance from a standing surface, the ATF position shall be determined in accordance with Figure 22.

No additional parameters or allowances shall be used for the MVP design option.



a_2	downwards installation tolerance for the conductors
a_4	downwards effect of conductor temperature
a_7	upwards installation tolerance for the conductors
a_8	upwards effects of conductor temperature
a_{11}	horizontal tolerance for the conductors
a_{12}	effects of wind on lateral position of conductor

Figure 22 – ATF design

The routing of the ATF at overline structures shall follow the hierarchy given in Table 25.

No.	Arrangement	Example
1	Bare feeder free running under the overline structure	Figure 30
2	Bare ATF fitted to the overline structure tensioned	Figure 31
3	Bare ATF fitted to the overline structure un-tensioned	Figure 32
4	Bare ATF oversailing the overline structure	Figure 33
5	High-level cabled ATF supported by a carrier wire free running through the overline structure	Figure 34
6	High-level cabled ATF supported from the overline structure	Figure 35
7	Low-level cabled ATF through troughing	Figure 36

Table 25 – ATF design arrangement hierarchy

The designer shall record the justification in the design decision log for each higher numbered preference (lower hierarchy) option not achieved before moving to the next option down (1 = most preferred) – see appendix C.

The ATF conductor height above a road surface shall be a minimum of 5.8 m.

11.1.1.2 Rationale

This hierarchy was developed as part of the ‘Application of Auto Transformer Feeding Design at Overbridges’ project (see module 3).

11.1.1.3 Guidance

The ATF minimum height above roadways is aligned to requirements in ENA TS 43-8 clause 6 and the Electricity Safety, Quality and Continuity Regulations 2002.

12 Electrical clearance control measures

12.1 General

In order to achieve the chosen electrical insulation categories, it may be necessary to provide control measures in accordance with the tables in module 5. For clarity these control measures are identified with a control measure (CM) number.

12.2 Minimum conductor cross-sectional area [CM01]

12.2.1.1 Requirement

Conductors shall be specified to achieve the minimum clearance listed in Table 26.

Conductor type	Minimum clearance	OCL	ATF
Solid conductors	0 mm	Yes	No
Stranded conductors with cross-sectional area ≥ 240mm (i.e. Centipede, Cockroach)	270 mm (dynamic)	N/A	Yes
Other stranded conductors	600 mm (static)	Yes	Yes

Table 26 – Minimum conductor cross-sectional area clearances

12.2.1.2 Rationale

This table is based on results from electrical short circuit arc testing NE-NR-0880-REP-7154 R02.

12.3 Insulated bridge arm [CM02]

12.3.1.1 Requirement

An insulated bridge arm shall be used to support the contact wires to achieve the electrical insulation category where required.

12.3.1.2 Rationale

The electrical performance of insulated bridge arms has been determined by Electrical testing is required in accordance with BS EN 50124-1:2017.

12.3.1.3 Guidance

Acceptable insulated bridge arms are:

Type	Product acceptance	UKMS drawing
Rebosio glass fibre bridge arm	PA05-06389	MS/B09/B01

Table 27 – Insulated bridge arm types

The Rebosio bride arm is 109 mm tall (when measured from the contact wire height). This is shorter than the historic bridge arm used by British Rail.

12.4 Elastic bridge arm [CM03]

12.4.1.1 Requirement

An elastic bridge arm shall be used to support the contact wires to achieve the electrical insulation category.

12.4.1.2 Rationale

The electrical performance of insulated bridge arms have been determined by Electrical testing is required in accordance with BS EN 50124-1:2017.

12.4.1.3 Guidance

Acceptable elastic bridge arms are per Table 28

Type	Product acceptance	UKMS drawing
Siemens elastic bridge arm	–	MS/B09/B02

Table 28 – Elastic bridge arm types

12.5 Water management measures [CM04]

12.5.1.1 Requirement

Water control measures such as guttering, or improvements to the construction of the overline structure, shall be installed if there is a likelihood of a stream of water or water jet falling onto live equipment.

12.6 Anti-roosting measures [CM05]

12.6.1.1 Requirement

Anti-roosting measures shall be installed if evidence of wildlife habitation in or near the structure is found.

Anti-roosting measures can include netting, bird slides, and other measures.

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12.7 Surge arresters [CM06]

12.7.1.1 Requirements

When a clearance is identified as basic (VCC), one surge arrester shall be connected on each side of the overline structure to each electrical section.

Surge arresters shall be situated as close to the structure as reasonably practical and less than 75 m from the overline structure face (see Figure 23).

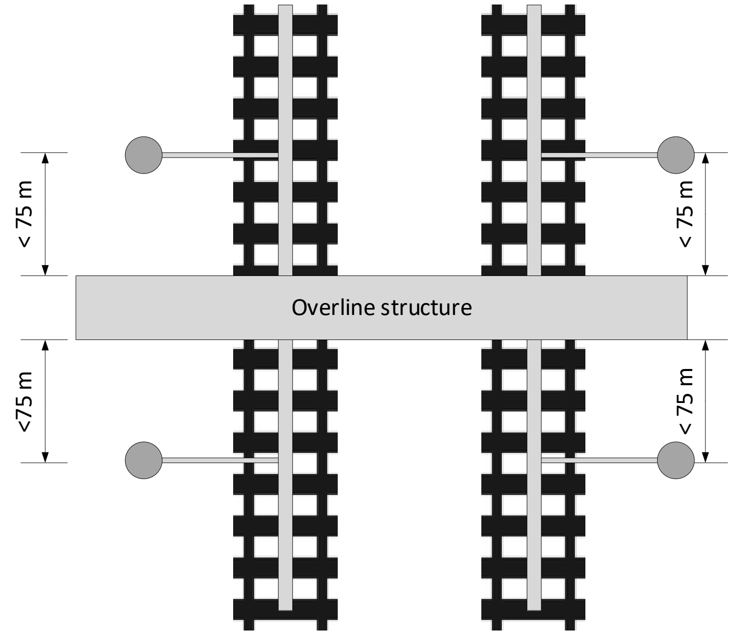

Figure 23 – Surge arrester locations

Surge arresters may be located in public areas.

Where a number of overline structures exist in close proximity, it is permissible to treat the installation as a system and reduce the number of surge arresters required.

If two adjacent overline structures are treated as a system then the maximum distance between surge arresters shall not exceed 450 m (see Figure 24) and the following shall not exist between surge arresters:

- neutral sections,
- in-line section insulators,
- on-load switches.

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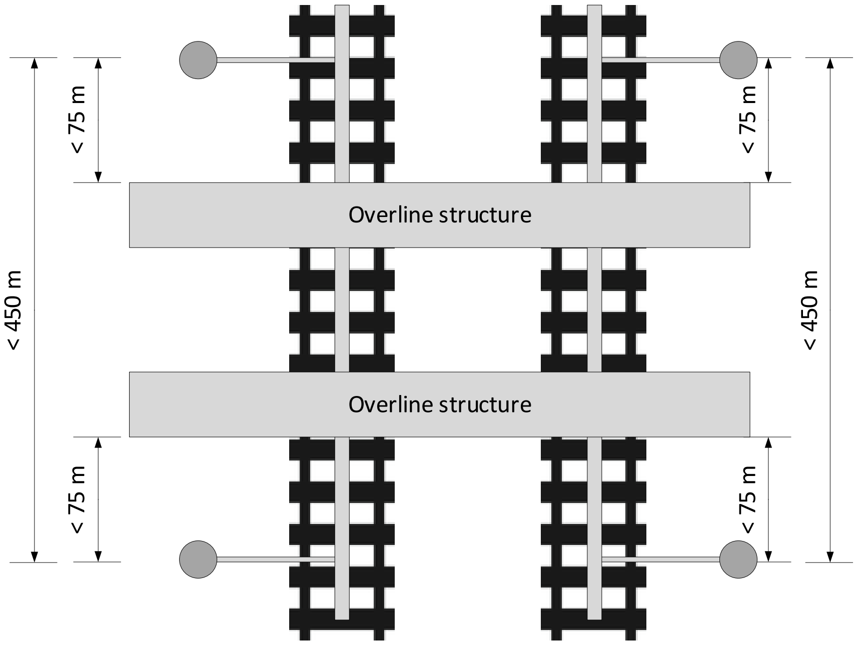

Figure 24 – Adjacent surge arresters

The position of the surge arrester on the OLE mast shall take into account the operation of the expulsion fuse and the drop-down earth jumper.

12.7.1.2 Rationale

The electrical performance of components when surge arresters are connected in circuit has been determined by Electrical testing is required in accordance with BS EN 50124-1:2017.

12.7.1.3 Guidance

The bonding of structures and surge arresters is detailed in NR/L2/ELP/21085.

NR/L2/ELP/27428 details the product specification for surge arresters.

The fitment of remote condition monitoring devices to surge arresters is optional.

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12.8 Contact wire covers [CM07]

12.8.1.1 Requirement

When an insulation category is identified as requiring contact wire covers these shall be fitted to the highest conductor under the bridge. The fitment of contact wire covers shall extend a minimum of 600 mm along track away from the overline structure (see Figure 25).

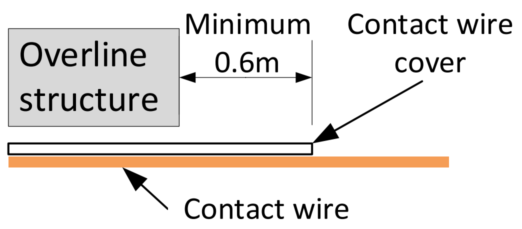

Figure 25 – Contact wire cover – along track coverage

The clearance from the cover shall be measured as shown in Figure 26.

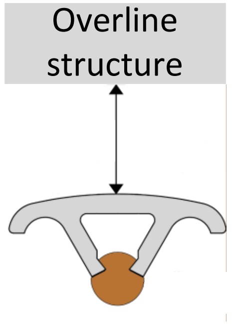

Figure 26 - Contact wire cover – Clearance measurement

12.8.1.2 Guidance

A single contact wire cover should be applied to the contenary at bridge approach spans.

Contact wire covers should be applied to the highest tensioned conductors in twin contact wire configurations (i.e. between bridge arms).

Covers designed for single or twin conductors are accepted for twin contact wire configurations.

Acceptable contact wire covers are:

- PA05/05084 – Kago OLE contact wire cover, or
- PA05/04984 – Furrer + Frey contact wire cover, or
- PA05/06529 – Horizon Insuline Birdguard.

12.9 Insulated coating [CM08, CM09]

12.9.1.1 Requirement

When an insulation category is identified as requiring an insulated coating, it shall be used to coat the overline structure to provide supplementary insulation. Two options exist for the application of insulated coatings (see Figure 27).

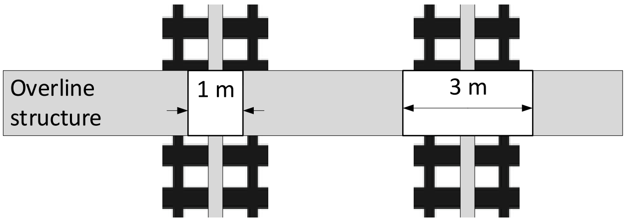

Figure 27 – Options for the application of insulated coating

12.9.1.2 Guidance

NR/GN/CIV/002 subclause 15.10 provides guidance on applying coatings for electrical insulation to infrastructure.

The wider coating strip provides an enhanced leakage path. It may also provide additional benefits for the clearance to pantographs. This is currently being quantified by high-voltage testing.

Acceptable insulated coatings are:

- GLS100R (minimum thickness of 4 mm).

12.10 Bonding

The requirement for flashover strips is detailed in NR/L2/ELP/21085.

12.11 Track datum plates

12.11.1.1 Requirement

Where a structure has clearance(s) categorised as basic (VCC), track datum plates shall be installed at the along-track positions coincident with all bridge arms.

12.11.1.2 Rationale

This is required to achieve managed track position tolerances.

12.11.1.3 Guidance

Track datum plates are specified in the following track standards:

- NR/L2/TRK/3201 – Management of tight clearances and track position,
- NR/L2/TRK/3417 – Specification, installation and maintenance of managed track position.

12.12 Track lateral resistance plates

12.12.1.1 Requirement

Where a structure has clearance(s) categorised as basic (VCC) and the track is of low fixity, the use of track lateral resistance plates should be considered.

12.12.1.2 Rationale

Track lateral resistance plates can help maintain the position of ballasted track.

12.12.1.3 Guidance

Track lateral resistance plates are fitted to sleepers or bearers to enhance lateral resistance to thermal and traffic forces.

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13 Track design

13.1 Tolerance of the track (a_1, a_5, a_{10})

13.1.1.1 Requirements

The horizontal and vertical track tolerances (a_1, a_5, a_{10}) shall include:

- track fixity – vertical;
- track fixity – horizontal;
- track fixity – cross level.

13.1.1.2 Rationale

The track tolerances applied in clearance assessments need to include all components of track fixity.

13.1.1.3 Guidance

The combined track tolerances are applied from a datum point midway between the two rail heads. This is shown in Figure 28.

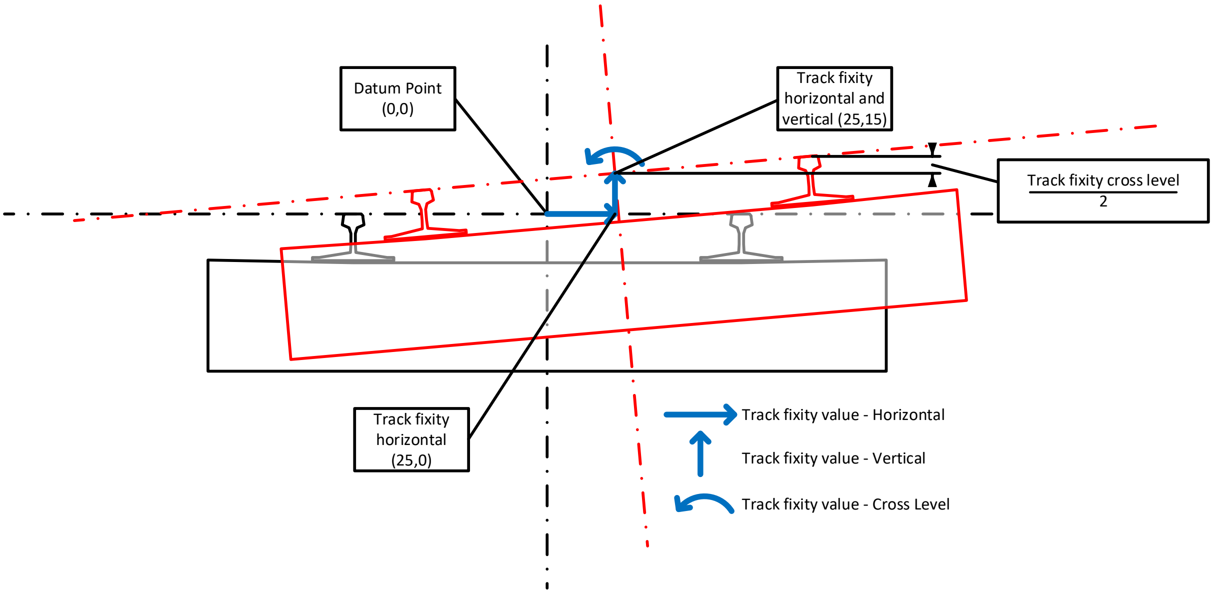


Figure 28 – Application of track fixity (not to scale)

13.2 Track – cant deficiency

13.2.1.1 General

Cant deficiency is the difference between the applied cant on the track and the equilibrium cant for the vehicle at the particular stated speed.

13.2.1.2 Requirement

Cant deficiency shall be determined for the track under the overline structure to determine the pantograph sway.

13.2.1.3 Rationale

Cant deficiency is required to determine the pantograph sway requirements detailed in GLRT1210, GIRT7073 and GMRT2173.

13.2.1.4 Guidance

Values

Cant deficiency is a calculated value and can be determined using equation below:

$$I_{max} = L.sin\left(arctan\left(\frac{V_{line}^2}{R_h.g}\right) - arctan\left(\frac{D - D_{tol}}{L}\right)\right)$$

I_{max} cant deficiency

L rail centres (1.435 m)

v_{line} speed (m/s)

R_h horizontal curve radius

g gravitational acceleration (9.81 m/s^2)

D installed cant

D_{tol} cross level error

(Source: GEGN8573 issue 4 G11.2.3.1.)

Probabilistic assumptions

No probabilistic assumption should be made for cant deficiency.

14 Pantograph

14.1 Pantograph outline

14.1.1.1 Requirements

The pantograph outline listed in Table 29 shall be used to determine clearances.

No.	Pantograph outline	Mechanically independent horn
1	BS EN 50367 Figure B.6	N
2	Brecknell Willis HS-P Mk2	Y
3	Brecknell Willis HS-X	Y

Table 29 – Pantograph outlines

Where modifications to existing electrification are planned, the clearances shall also be assessed using the pantograph outline from BS EN 50367 Figure B.6.

14.1.1.2 Rationale

RIS-8270-RST subclause 3.4 requires the proposer to produce the criteria by which technical compatibility at route level can be determined.

14.1.1.3 Guidance

Module 4 of this manual contains further details of each pantograph outline.

Network Rail, with industry partners, is currently undertaking in-service trials of insulated pantograph horns. These capitalise on the design of modern pantographs with independently suspended parts of the pantograph head.

RIS-8270-RST details the process for producing a statement of compatibility.

14.2 Pantograph head profile – static

14.2.1.1 Requirements

The pantograph head profile – static shall be determined using the pantograph outline dimensions (horn & carbon carriers). No effect from contact force on secondary suspension shall be included.

14.2.1.2 Rationale

This provides a pantograph position for trains when assessing static clearances.

14.2.1.3 Guidance

Pantograph horns could be insulated, and therefore the carbon carrier and horns need to be separately identified in the profile.

14.3 Pantograph head profile – active

14.3.1.1 Requirements

The pantograph head profile – active shall be determined using the following factors.

- pantograph outline dimensions (horn & carbon carriers);
- head roll;
- carbon wear; and
- pantograph sway

14.3.1.2 Rationale

This provides a pantograph profile for moving trains when assessing dynamic clearances.

14.3.1.3 Guidance

Pantograph horns could be insulated, and therefore the carbon carrier and horns need to be separately identified in the profile.

14.4 Pantograph head profile – active (probabilistic)

14.4.1.1 Requirement

The pantograph head profile – active (probabilistic) shall be determined using the following factors.

- pantograph outline dimensions (horn & carbon carriers);
- head roll;
- carbon wear;
- pantograph sway.

When applying probabilistic pantograph profiles, the calculated clearances shall be categorised in accordance with Table 30.

Clearance category	**Standard deviations (SD)**
Average clearance	0 SD
Typical clearance	1 SD
Exceptional clearance	3 SD

Table 30 – Probabilistic classification

Probabilistic pantograph profiles shall only be calculated using probabilistic values for the key parameters as detailed in module 4. Additional probabilistic assumptions shall not be used.

When using probabilistic techniques, a minimum of 10,000 iterations shall be calculated to determine the clearances.

14.4.1.2 Rationale

The principles of probabilistic assessment of the pantograph profiles are detailed in the RSSB research report 'Probabilistic Gauging', which is part of project T1750 – Avoiding bridge reconstruction.

Probabilistic gauging can also be applied to flat deck overline structures, but the benefits are limited.

10,000 iterations was recommended by the T1750 project – RPT001.

14.4.1.3 Guidance

Probabilistic clearance categories "average" and "typical" provide additional information to support engineering decisions.

15 Obstacles (including screens)

15.1.1.1 General

This clause will be moved to module 2 and enhanced in a future issue of this standard.

15.1.1.2 Requirement

Where clearances set out in subclauses 8.14, 8.15, 8.16 and 8.20 of this module cannot be achieved, protection by obstacle shall be provided as protection against direct contact with live parts, in accordance with BS EN 50122-1:2022 subclauses 5.3.1 to 5.3.5.

Bridge parapets

The requirements for the bridge parapet shall be assessed using NR/L3/ELP/27717.

Signal screens

Signal structure working platforms shall be assessed as a restricted non-public area.

Where the restricted non-public area clearance is not achieved, the mitigations in Table 31 shall be considered.

No.	Mitigation
1	Install signage to remind staff that signals are a restricted non-public area, or install ladder locks, or remove legacy ladders
2	Install screen to existing signal
3	Replace existing signal head with a maintenance-free or fold down type
4	Install new signal

Table 31 – Signal clearance mitigations

Option 1 shall be considered as the MVP design option.

The designer shall identify options to inform the project funder of opportunities to optimise mitigations (other than ladder locks). To provide transparency to funders, these options shall be treated as additions to the MVP.

A risk assessment for working closer than 2.75 m shall be completed by the designer for the Electrical Power Maintenance Engineer (EPME) to review and modify if required – see appendix D.

15.1.1.3 Rationale

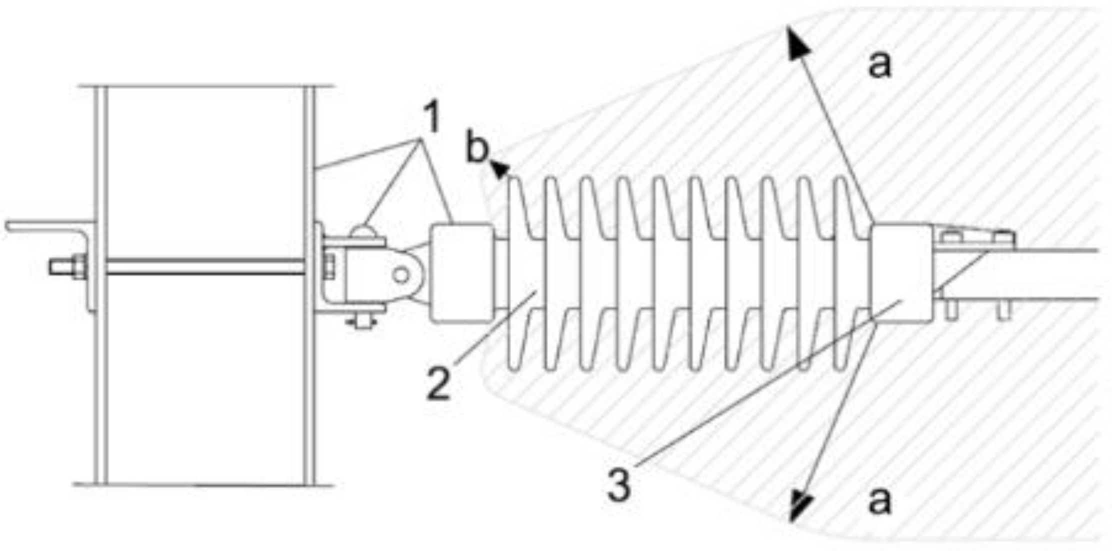
A risk assessment is required by NR/L3/ELP/29987 for working closer than 2.75 m to live equipment.

15.1.1.4 Guidance

Signal maintenance positions should be designed to maximise, so far as is reasonably practicable, the electrical clearances from the signal infrastructure, and personnel accessing it, to the OCLS and live parts of the pantograph.

An example of an EPME risk assessment is contained in Appendix D.

Appendix A – Determining electrical clearance to insulators



Key

1 earthed equipment

2 insulator sheds

3 live equipment

a value of electrical clearance

b 10 mm

Figure 29 – Electrical clearances to insulators (extract from BS EN 50119:2020 figure 2)

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Appendix B – ATF overline structure arrangements

B.1 Bare feeder free running under the overline structure

This is the simplest solution for routing the ATF conductor under an overline structure. Structures either side of the overline structure support the ATF at a height that allows the conductor to span through the structure maintaining all electrical clearances.

To achieve the required clearances, the overline structure needs to have both vertical headroom and lateral space.

An example is shown in Figure 30.

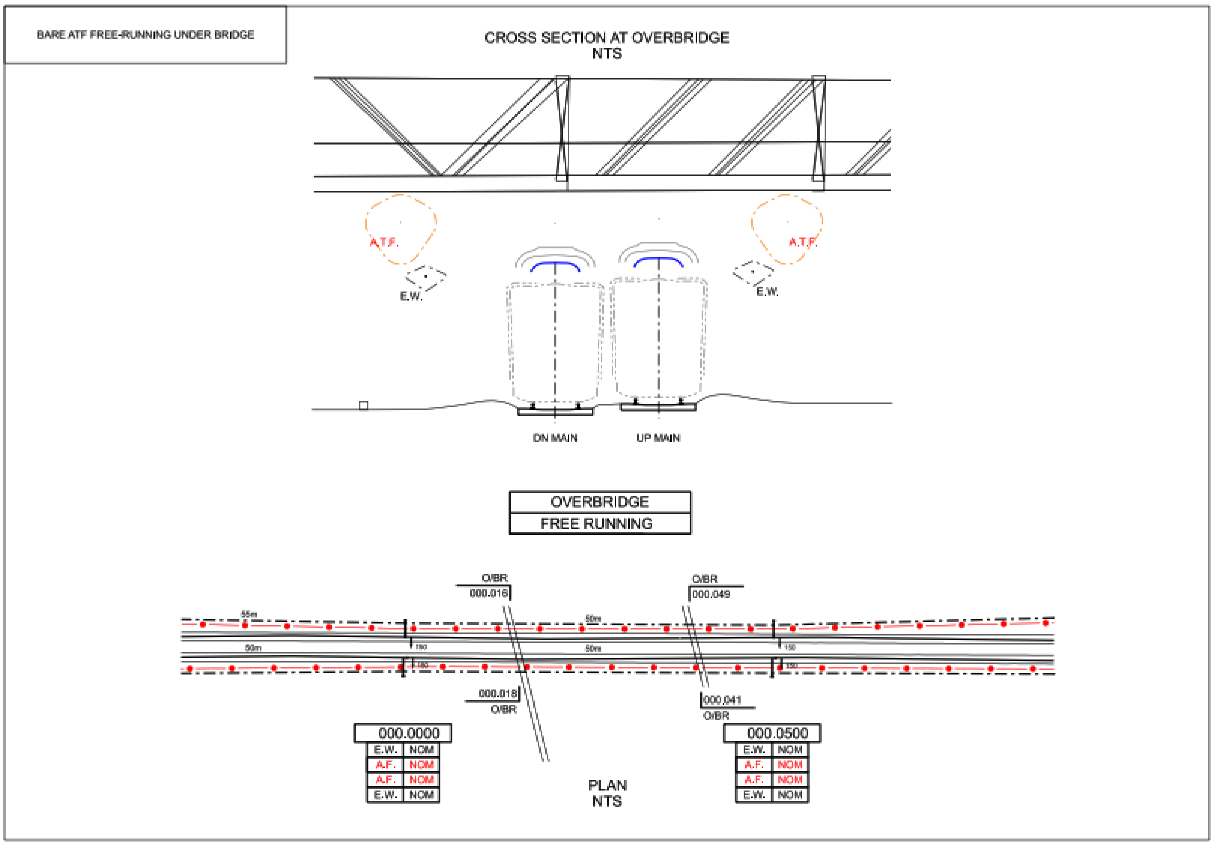

Figure 30 – Example of bare feeder free running under an overline structure

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B.2 Bare ATF fitted to the overline structure tensioned

ATF insulator support assemblies are installed on the overline structure walls. The ATF conductor is supported by the insulator support assemblies allowing the ATF to go through the structure tensioned.

An example is shown in Figure 31.

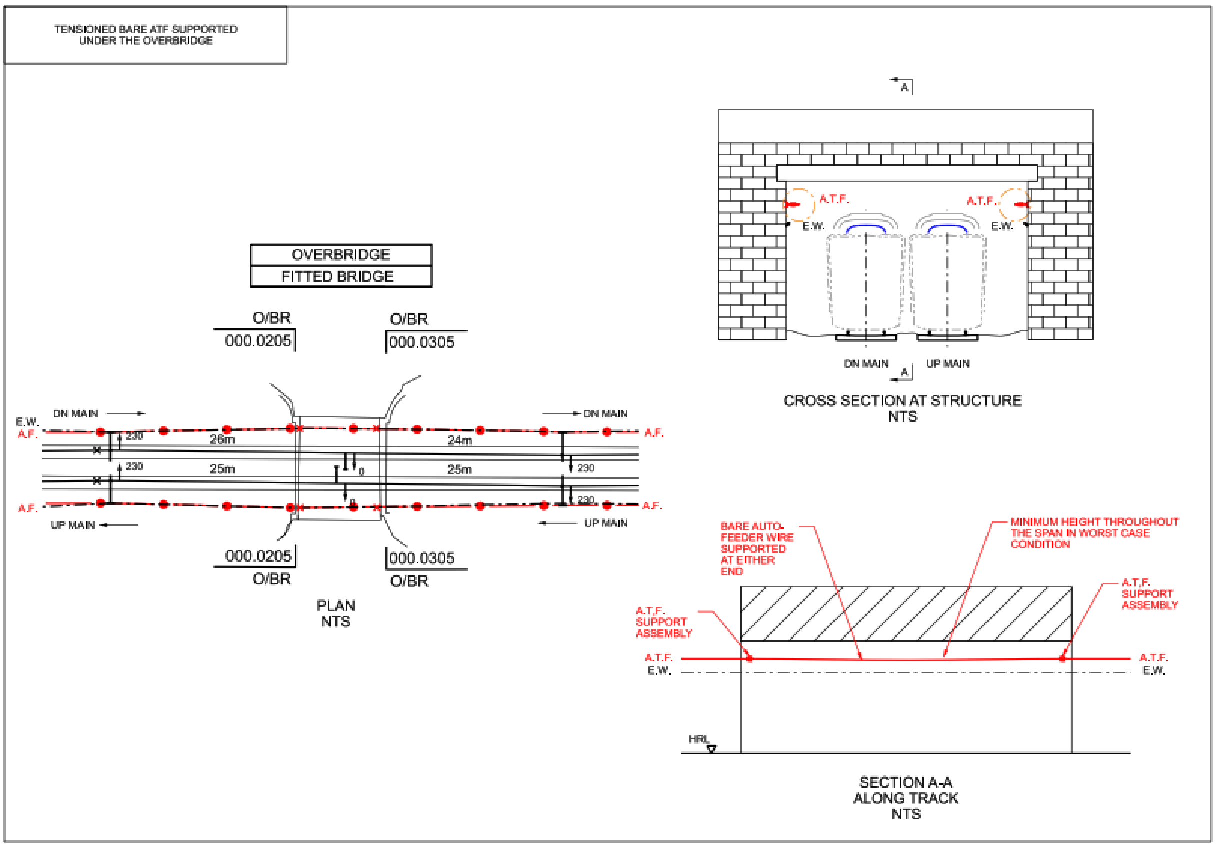

Figure 31 – Example of bare ATF fitted to the overline structure tensioned

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B.3 Bare ATF fitted to the overline structure un-tensioned

The ATF is terminated on either side of the overline structure with a jumper to an aluminium rod supported by an insulator on the structure wall. The ATF conductor is electrically connected to the aluminium rods at each structure face and supported through the structure by insulators attached to the wall.

An example is shown in Figure 32.

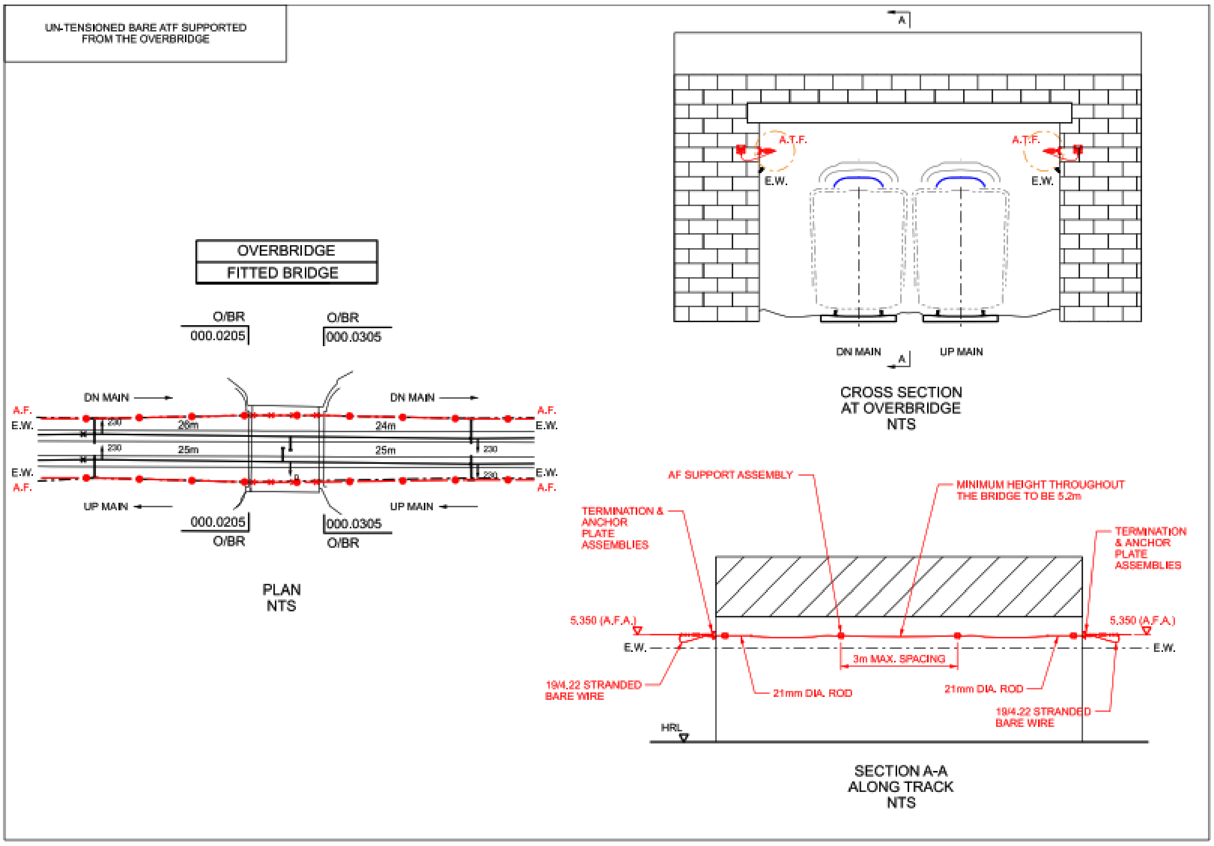

Figure 32 – Example of bare ATF fitted to the overline structure un-tensioned

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B.4 Bare ATF oversailing the overline structure

B.4.1.1.1 General

The ATF conductor is raised along the railway to enable a minimum height when crossing the road and then to be lowered to its normal height along the railway.

B.4.1.1.2 Requirement

The ATF conductor height above the road surface shall be a minimum of 5.8 m.

An example is shown in Figure 33.

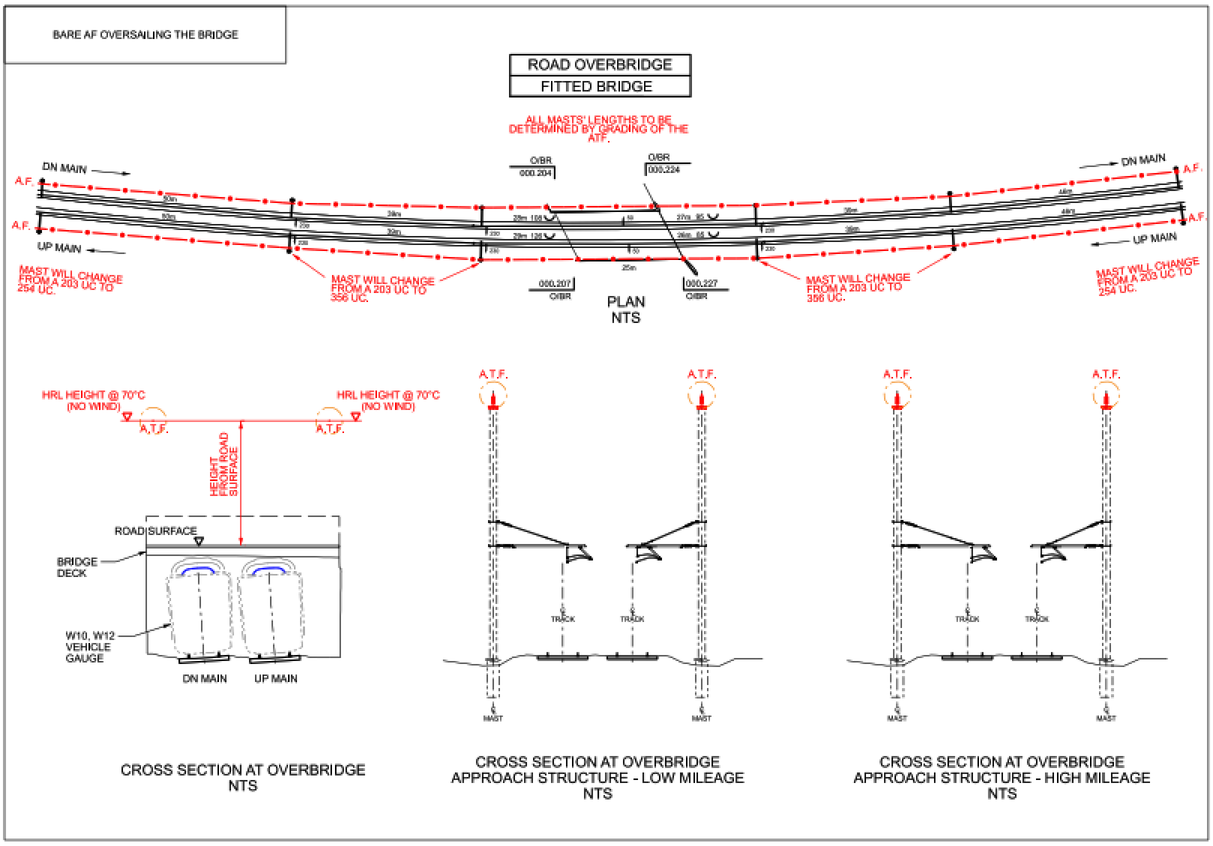

Figure 33 – Example of bare ATF oversailing the overline structure

B.4.1.1.3 Rationale

This arrangement meets the requirements of:

- ENA TS 43-8 clause 6,
- The Electricity Safety, Quality and Continuity Regulations 2002,
- The Highways Act 1980, section 178, para (5),
- New Roads and Street Works Act 1991.

B.4.1.1.4 Guidance

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Network Rail is a statutory undertaker responsible for maintaining and operating the country's railway infrastructure.

The maximum UKMS OCLS mast height is 15 m.

B.5 High-level cabled ATF supported by a carrier wire free running through the overline structure

The live ATF conductor is anchored on a mast that is either a self-supporting anchor or a tie-wire anchor. The overline structure side of the mast has a carrier wire anchored (normally the same conductor as the ATF) that routes through the overline structure to a mast the other side. The insulated feeder cable is connected through a sealing end to the live ATF and then supported by the carrier wire through the overline structure. The carrier wire anchor height can vary on the mast to suit the headroom of the bridge.

There is a tension difference between the ATF anchor and the carrier wire anchor due to the vertical load difference; for this reason the overline structure faces are required to be anchors.

An example is shown in Figure 34.

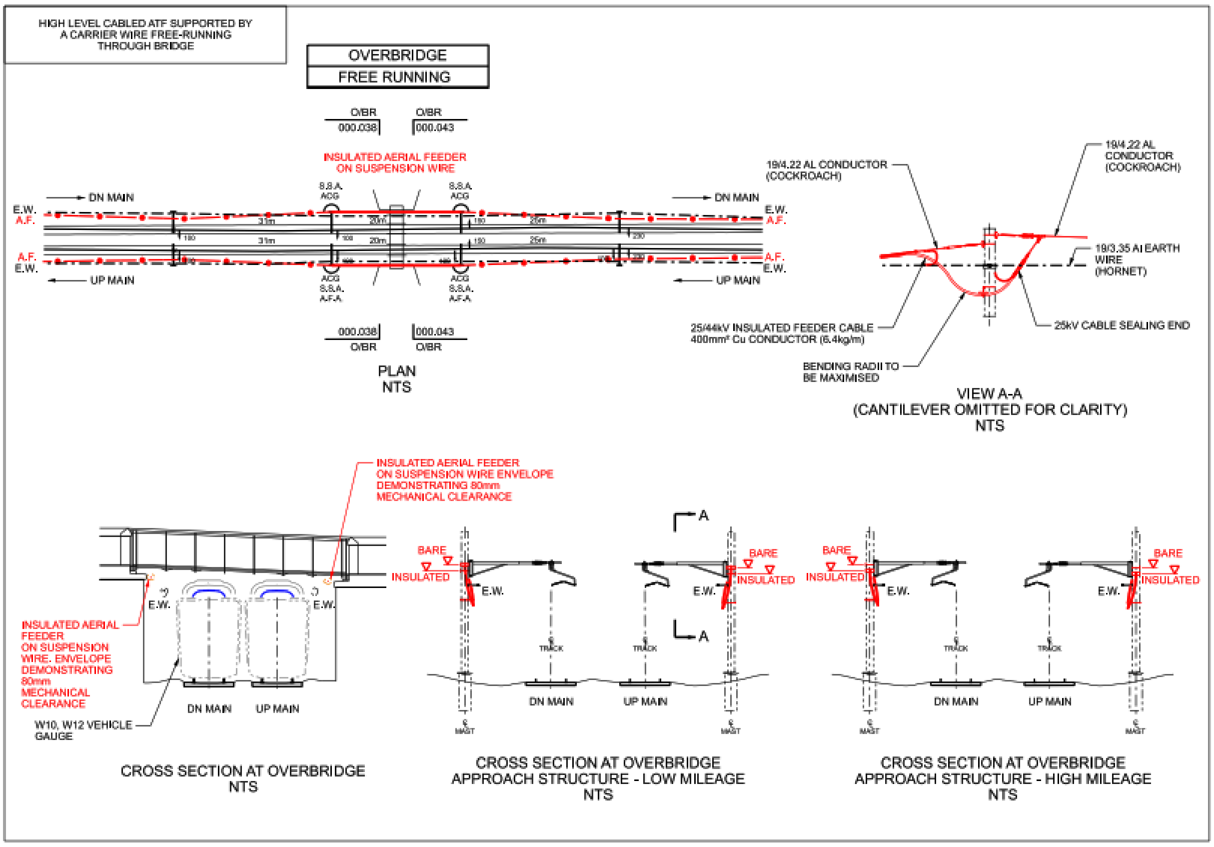

Figure 34 – Example of high-level cabled ATF supported by a carrier wire running through an overline structure

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B.6 High-level cabled ATF supported from the overline structure

The bare ATF is terminated on an overline structure face that is either an anchor tied mast or a self-supporting mast. An earthed conductor is strung between the overline structure face mast and the face of the overline structure under nominal tension. The cabled ATF is connected through a sealing end to the bare ATF and then supported by the earthed conductor. It is then routed through the overline structure and supported at 2.00 m nominal spacings from the wall of the overline structure.

An example is shown in Figure 35.

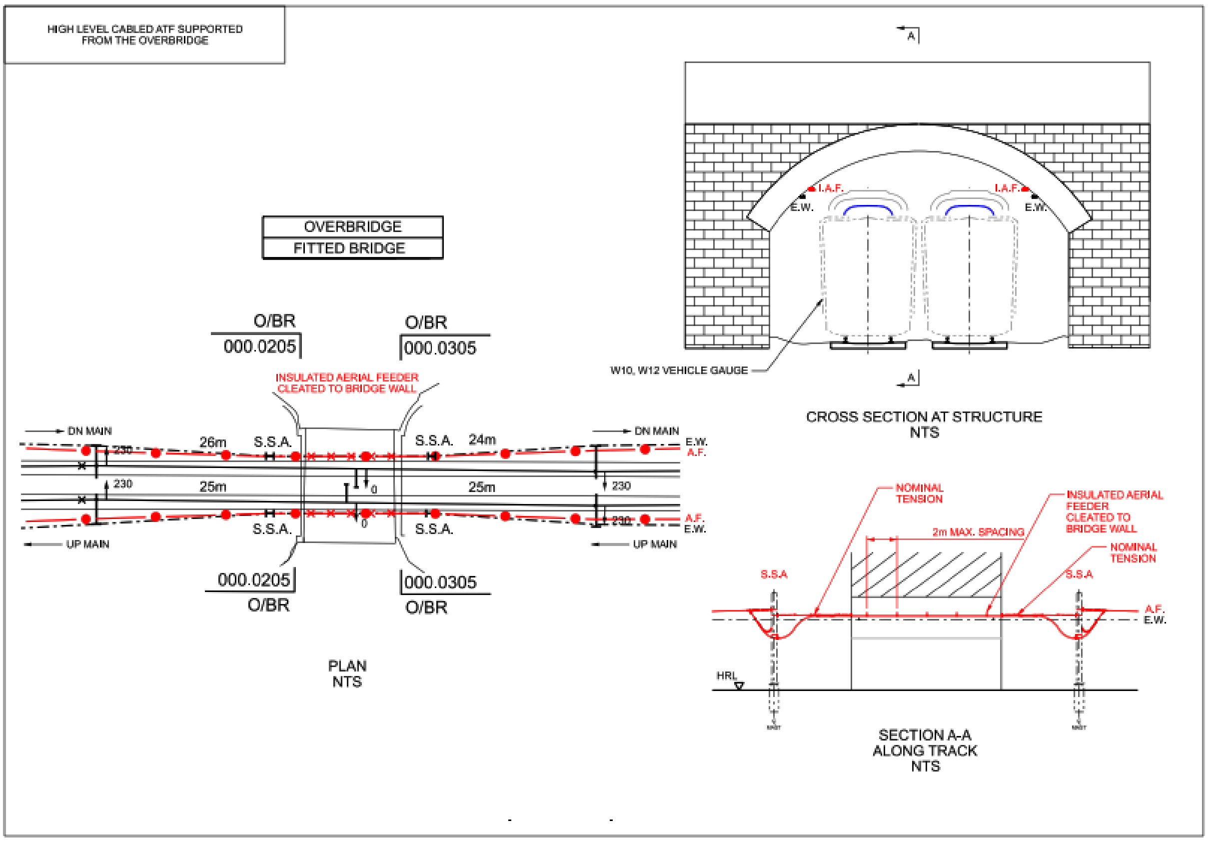

Figure 35 – Example of high-level cabled ATF supported from the overline structure

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B.7 Low-level cabled ATF through troughing

This design terminates the ATF bare conductor on overline structure face and then cables the gap. The cable is connected to the bare ATF through a sealing end and is then routed down the mast to troughs at ground level through the bridge and then back up to the bare ATF conductor through sealing ends.

An example is shown in Figure 36.

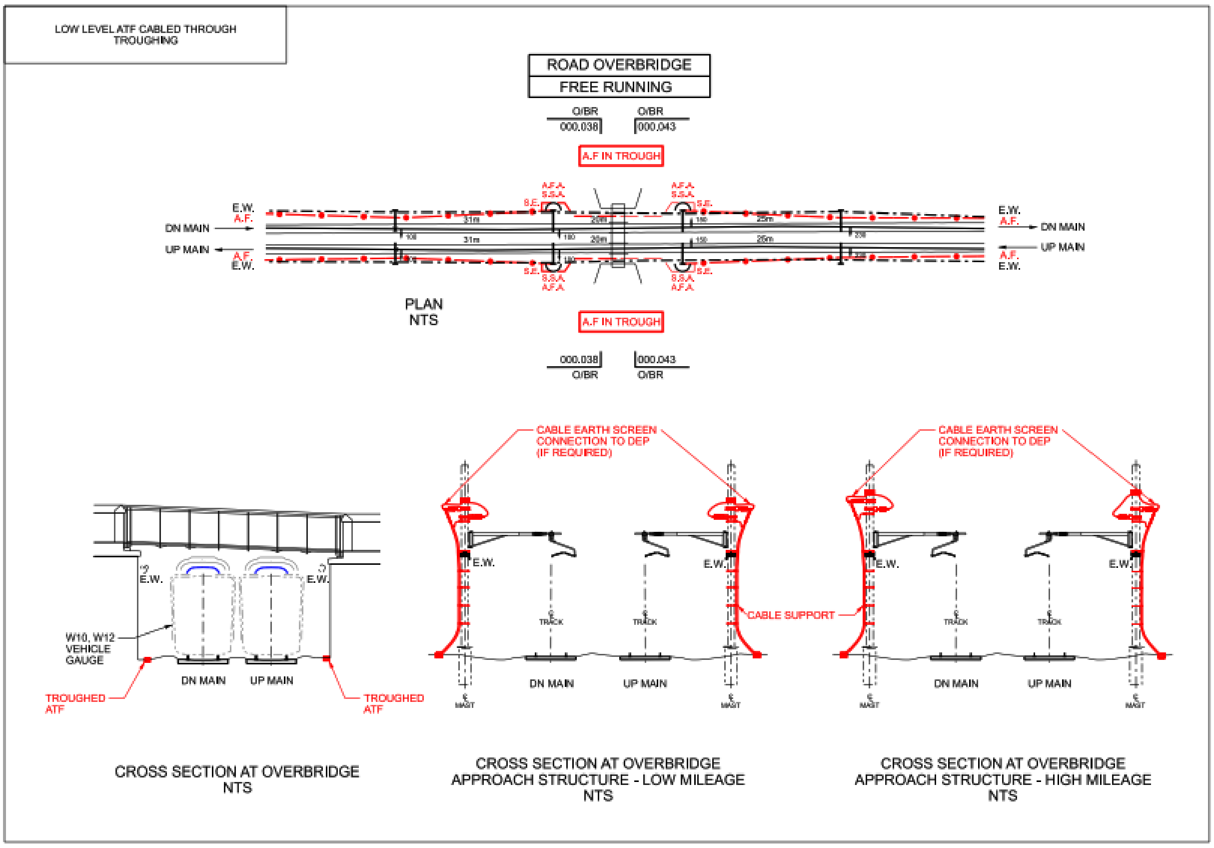

Figure 36 – Example of low-level cabled ATF through troughing

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Appendix C – Design decision log

C.1 Section 1 – Overline Structure Information

Overline Structure - Electrical Clearance Risk Assessment

Structure Details							
Structure Name							
Structure Number							
Type of Structure							
Description of site / structure							
Structure owner							
Listed status							
Bridge construction material							
Date of last structure examination (NR/L3/CIV/006)							
Are there concerns from the Management of Structures Manual (NR/L2/CIV/032)							
Services in structure							
Location							
Region		ELR		Mileage from		Mileage to	
Hazard relevant to electrification identified from structure examination							
Water							
Birds							
Other							
Proximity of other infrastructure							
Stations							
Other Structure							
Foot crossings							
Level crossings							
Standing surfaces							
Credible standing surfaces							
Drainage							

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C.2 OCLS Electrical Clearances (per road) part 1

Overline Structure - Electrical Clearance Risk Assessment

Structure Details

Structure Name	
Structure Number	
Type of Structure	
Track / Road	

Clearance

	Permitted arrangements								
	Insulation categories				OCLS arrangement			Contact wire height	
Number	Reinforced	Basic	Functional	Basic (VCC)	Free running	Reduced system depth	Twin contact	< 4165 mm	=≥ 4165 mm
1	✓				✓			✓	
2		✓			✓			✓	
3	✓					✓		✓	
4			✓		✓			✓	
5	✓						✓	✓	
6		✓				✓		✓	
7		✓					✓	✓	
8			✓				✓	✓	
9				✓			✓	✓	✓

	EC01		EC02		EC03		EC04		EC05		EC06		EC07	
	OCLS conductors (including contact wire)		OCLS conductors (including contact wire)		OCLS conductors (including contact wire)		Support (including underbridge arm)		Support (including underbridge arm)		Pantographs – head		Pantographs – horn	
	Rail vehicle bodies		Overline structures (bridges, tunnels, canopies)		Overline structures (bridges, tunnels, canopies)		Overline structures (bridges, tunnels, canopies)		Overline structures (bridges, tunnels, canopies)		Overline structures (bridges, tunnels, canopies)		Overline structures (bridges, tunnels, canopies)	
	–		Static		Dynamic		Static		Dynamic		Dynamic		Dynamic	
Number	Dimension [mm]	Insulation category	Dimension [mm]	Insulation category	Dimension [mm]	Insulation category	Dimension [mm]	Insulation category	Dimension [mm]	Insulation category	Dimension [mm]	Insulation category	Dimension [mm]	Insulation category
1														
2														
3														
4														
5														
6														
7														
8														
9														

C.3 OCLS Electrical Clearances (per road) part 2

Decision

Control Measures (MVP)				Additional Control Measures	DPE Agreement	Sponsor Agreement
CM01	Min. Conductor CSA					
CM02	Insulated Bridge Arm					
CM03	Elastic Bridge Arm					
CM04	Water management measures					
CM05	Anti Bird measures					
CM06	2 * Surge Arrestors					
CM07	Contact Wire Cover					
CM08	Insulated Coating (1m wide strip)					
CM09	Insulated Coating (3m wide strip)					
CM10	Insulated Pantograph Horn					
Low Wire Height Mitigation						
1	Included on isolation diagrams as a residual hazard					
2	Install onsite signage using RTE 6029 warning sign					
3	Include on the route OCLS list of locations affected by emissions from a steam locomotive					
4	Publish in the route hazard record					

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C.4 ATF Electrical Clearances (per electrical section)

Overline Structure - Electrical Clearance Risk Assessment

Structure Details	
Structure Name	
Structure Number	
Type of Structure	
ATF Adjacent to track / road	

ATF Hierarchy		Selected / Reject (including reason for rejection)
1	Bare feeder free running under the overline structure	
2	Bare ATF fitted to the overline structure tensioned	
3	Bare ATF fitted to the overline structure un-tensioned	
4	Bare ATF oversailing the overline structure	
5	High level cabled ATF supported by a carrier wire free running through the overline	
6	High level cabled ATF supported from the overline structure	
7	Low level cabled ATF through troughing	

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Appendix D – Access to signalling – risk assessment

Our Lifesaving Rules

Validity	30-07-20 to 30-07-21

E&P Task Briefing

Briefing Title	Access to Signal WN178 WN178 (Royal Mail)
Location Golborne Junction	
ELR	CGJ-3
Mileage	183m 68ch
This Task Brief is to be read in conjunction with standard S&T Role Call requirements	

Rev	Prepared by	Date	*Approved by
			E&PME

A Task Details:
Warrington Area Signal Gantry Risk Assessment for encroaching 2.75m Electrical Clearance during Signal Climbing Activities

A1 Description of the Work:
The COSS will be in control of the site and will ensure that he/she is aware of all on site by ensuring that the Staff sign the Safe System of Work Pack (SSOW) after receiving and signing the appropriate task briefing.
Staff can be allowed to climb the signal structure and encroach the 9ft (2.75m) electrical clearance in line with the "Green Book".
NR/L3/ELP/29987 'working on or about 25kV AC electrified lines', permits access to a signal when a person is less than 2.75m from live parts of OLE, provided the signal and work activity has been risk assessed and adequate control measures are put in place to prevent persons, tools or equipment coming within 600mm of live parts of overhead line equipment.

WN178, WN179
Final solution is to fit screening to side of ladder, on both ladders.
New screening fitted at top of gantry walkway sides.
Hooped ladders both sides of track.
1.319m from ladder hoop to ATF near top of ladder.
Ladder hoop is safe distance demarcation, person, tools and equipment must not reach outside of ladder hoops.
Can be accessed without isolation.
Bond must be confirmed to be present and connected.
PIC to observe and ensure clearances are maintained

A2 Control of Activity Risks (H&S and Env)

29987 Module 2 extract

4.3 Work Less Than 2.75 metres (9 feet) but More Than 600 millimetres (2 feet) from Live Equipment (See Appendix A)

When work is to be attempted which can, or is LIKELY to, result in any part of a person's body or clothing or anything to be used:

a) coming within 2.75 metres (9 feet) but NOT within 600 millimetres (2 feet) of any live part of the OLE; and/or

b) within the vertically unbounded space above 2.75 metres to either side of the live OLE; and/or

c) within 2.75 metres but NOT within 600 millimetres of live pantographs and other roof-mounted electrical equipment on trains.

A risk assessment shall be undertaken and a documented safe system of work shall be prepared by a competent person in accordance with Module 3.

29987 Module 3 Extract.

3.3 Work Less Than 2.75 metres (9 feet) but More Than 600 millimetres (2 feet) from Live Equipment (See Appendix A of Module 2)
The competent person preparing the documented safe system of work required by Module 2 shall describe in the safe system of work how it is intended to carry out the work without coming within 600 millimetres (2 feet) of any live part of the OLE, or live pantographs and other roof-mounted electrical equipment on trains.
No work or any part of the body shall come within 600 millimetres (2 feet) of live equipment.
The competent person shall include in the safe system of work:
a) a description of the access to, nature and situation of the proposed work; **and**
b) the tools, materials, plant, appliances and liquids to be used; **and**
c) the details of any isolation considered necessary; **and**
d) the competence and responsibilities of the persons involved in the proposed work; **and**
e) the control measures proposed to be taken.

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Risk	Control Measures
Access – Unauthorised access by members of the public	Clear and adequate signage. Maintain fenced off worksites. Keep all access points secured.
Electrified Lines (OLE) 25KV	Compliance and briefings to GE/RT8000. Adherence to NR/SP/ELP/29987, working instructions for AC electrified lines. Use of insulated tools and equipment
Poor Ground Condition	Check PCIP & Hazard directory. Interface with Network Rail Maintenance Good house keeping Use of CAT scanning equipment Trained and competent operators Use of authorised walking routes.
Mobile Phone Use.	Restrictions on use of mobile phones to be briefed in site inductions, specific task briefings and work package planning. Use of mobile phones on site to be minimised to authorised key personnel. Authorised Persons to restrict use in position of safety only. Include in task briefings and site induction procedure.
Site Hazards – Health Hazards	Site inductions to include procedure for dealing with the safe removal of sharps. Good housekeeping standards Waste management plan in place Correct selection and use of PPE.
Manual Handling	Use of mechanical handling aids Manual handling risk assessments Trained and competent operatives Site inductions
	Specific task briefings and work package planning.
Darkness – Work in the hours of darkness	Site induction Clearly defined work sites Specific task briefings Provision of 110v lighting and back up generators. VT Tower Lights
Slips Trips & Falls	Good housekeeping Effective waste management planning Specific task briefings Site induction to include waste management plan.

3 Resources
Standard PPE.
All Orange Hi-Vis Clothing, Lace Up Safety Boots, Safety glasses, Gloves, Hard Hat, Safety Harness
Task Specific PPE
Ladder Working requires Rescue and Recovery equipment
Labour
The following competent (certificated) resources will be on site:-

- Coss / PIC

B4 Protection Arrangements
Site Warden Warning

6 Communications and Contact Details:

Contact	Name	Tel: Number
S&TME		
S&T SM		
S&T SS		
E&PME		

B7 Emergency Arrangements
In the event of the emergency services being required they must be directed to the point nearest the accident:-

- Address DALLAM Royal Mail, WA5 0LZ
- Grid Ref: SJ 60363 89022

All accidents and incidents shall be reported to the LNW DU On Call S&T Manager who shall advise the RRC and NDS 24/7
Following any accident or incident work shall be stopped. Only the Site Supervisor / On Call Manager shall authorise the re-start.

Contact	Name or Location	Tel. Number
Nearest 24 Hr A&E Hospital **999 in Emergency**		
Signaller		
NR RRC Manager		
NR Close call System		
ECO		
BT Police		

B8 Welfare:
NWR welfare facilities are available at Warrington DU site, with as a minimum: Hot water, Toilet and Microwave

First Aid Arrangements
First aiders shall be identified during the COSS Briefing.
First aid kits shall be located at:

- Worksite
- Local DU Welfare facility
- PSB / Stations and Site Offices (Where available)

C Briefing:

C1 Briefing Record:
A record of the Task Brief shall be maintained. No person shall be allowed to work unless they understand and have signed the briefing.
C2 Changes to a Task Brief
Changes and the reason for the change shall be recorded. Changes shall be reviewed against sections A2 and B3. The change shall be approved by the Responsible Manager.

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Our Lifesaving Rules

Lancs & Cumbria E&P Task Briefing Record

Task Brief Title and Reference	Access to Signal WN178, WN179
Project Title and Reference	Warrington Area Signal Gantry Risk Assessment for Encroaching 2.75m Electrical Clearance during Climbing Activities

Briefing Given By:

Name	Sentinel Number	Position	Signature

Briefed To: (By signing below, I confirm that I have received and understood the briefing for this task. I am also declaring my fitness to work in accordance with Network Rail's Drug and Alcohol and Fatigue Management policies.)

Name	Sentinel Number	Signature	Date & Time

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